Protograph LDPC-Coded Non-Uniform SCMA With Segment Wise Interleaved Transmission: A Promising Coded-Modulation Technique for SWIPT-Enabled 6G Networks

Zhaojie Yang, Yunye Li, *Graduate Student Member, IEEE*, Xiaoxi Yu, Jiayi Yang, Yi Fang, *Senior Member, IEEE*, Yong Liang Guan, *Senior Member, IEEE*, and Yuhao Chi, *Senior Member, IEEE*

***Abstract*—This paper is concerned with sparse code multiple access (SCMA) with protograph-based low-density parity-check (P-LDPC) codes over Rayleigh fading channels. To be specific, we first propose a two-step design method consisting of both low-density factor graph (LDFG) and energy-based constellation superposition (ECS) principles to construct a new type of non-uniform codebooks (CBs), which have superior robustness against inter-user interference. We also develop a new segment-wise interleaved transmission (SWIT) scheme, which features a systematic optimization among repetition coding, resource mapping, and cross-slot data coupling, to further improve the performance of P-LDPC-coded SCMA. Additionally, we conceive a decoding-threshold-guided multi-objective optimization (DT-MOO) method to construct protograph-based enhanced multi-user codes (EMUCs), which exhibit excellent error-rate performance under both turbo and non-turbo decoding compared to traditional single-user and existing multi-user codes. Theoretical and simulated results validate the superiorities of proposed non-uniform SCMA scheme with SWIT and EMUCs compared to state-of-the-art benchmarks. Owing to the above advantages, the proposed designs are competent to provide massive connectivity and energy-efficient transmission for simultaneous wireless information and power transfer (SWIPT) applications in 6G networks.**

***Index Terms*—Massive connectivity, sparse code multiple access (SCMA), constellation superposition, coded modulation, multi-objective optimization, 6G network.**

I. Introduction

A. Background

SIMULTANEOUS wireless information and power transfer (SWIPT) [1] can coordinate the transmission of information and power over a single radio frequency (RF) resource, providing a feasible solution to remotely charge low-energy devices. The SWIPT is suited for satellite communications [2], covert communications [3], integrated sensing and communications [4], and 6G Internet of Things (IoT) networks [5]. In particular, SWIPT-enabled IoT facilitates a fundamental shift in communication networks from a human-centric paradigm to massive machine-type communication (mMTC), which is critical for 5G-and-beyond applications [6]. In this context, the traditional orthogonal multiple access (OMA) struggles to support massive connectivity due to its inherently limited resource efficiency. To increase the number of supported devices, non-orthogonal multiple access (NOMA) [7] has been proposed as a revolutionary technology allowing multiple user equipments (UEs) to share the same time-frequency resource elements (REs). NOMA techniques can be typically categorized into power-domain NOMA (PD-NOMA) [8] and code-domain NOMA (CD-NOMA) [9]. The former and latter consider the superposition of user signals generated from different power levels and specially structured codebooks (CBs), respectively, enabling multi-user access on an identical RE.

A representative technique in CD-NOMA is called *sparse code multiple access (SCMA)* [10], where each UE with a pre-designed sparse CB maps the binary bits into multi-dimensional complex modulation symbols. Such a bit-to-symbol mapping mechanism not only enables a sparse superposition of multiple user signals over the shared REs (i.e., a sparse graph model), but also facilitates the multi-user detection at receiver via message passing algorithm (MPA) [11]. Nevertheless, due to the limited REs accessed by multiple UEs, inter-user interference is inevitable and will intensify as the system overload increases, which becomes a primary

Received 14 October 2025; revised 18 March 2026; accepted 28 May 2026. Date of publication 8 June 2026; date of current version 12 June 2026. The work of Zhaojie Yang was supported in part by the NSF of China under Grant 62501173; in part by the National Research Foundation, Singapore and Infocomm Media Development Authority under its Future Communications Research & Development Programme under Grant FCP-NTU-RG-2022-020. The work of Yi Fang was supported in part by the NSF of China under Grant 62322106 and Grant 62571142 and in part by the International Collaborative Research Program of Guangdong Science and Technology Department under Grant 2025A0505020081. The work of Yuhao Chi was supported in part by NSFC under Grant 62571399 and Grant 62471357 and in part by the Fundamental Research Funds for the Central Universities under Grant QTZX26126. *(Corresponding author: Yi Fang.)*

Zhaojie Yang, Yunye Li, and Yi Fang are with the School of Information Engineering, Guangdong University of Technology, Guangzhou 510006, China (e-mail: yangzhaojie@gdut.edu.cn; huohuasysu@hotmail.com; fangyi@gdut.edu.cn).

Xiaoxi Yu and Yong Liang Guan are with the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore 639798 (e-mail: xiaoxiyu@ntu.edu.sg; eylguan@ntu.edu.sg).

Jiayi Yang is with the School of Computer Science and Engineering, Sun Yat-sen University, Guangzhou 510006, China (e-mail: yangjy256@mail2.sysu.edu.cn).

Yuhao Chi is with the State Key Laboratory of Integrated Services Networks, Xidian University, Xi'an 710071, China (e-mail: yhchi@xidian.edu.cn).

Digital Object Identifier 10.1109/JSAC.2026.3700994

bottleneck to SCMA performance. Thereby, the inter-user-interference-resistant techniques, including CBs [12], [13], [14], [15], [16], [17], [18], [19], repetition coding [20], [21], [22], [23], [24], and error-correction codes [25], [26], [27], [28], [29], have attracted considerable research attention from academia.

B. Related Works

Most CBs are constructed from a pre-defined multi-dimensional mother constellation with phase rotation. An example is the CB proposed by Huawei [12], whose multi-dimensional constellations for different UEs are obtained by rotating the mother constellation. Note that to generate a good CB, a large minimum Euclidean distance (MED) for multi-dimensional constellations is crucial, which has motivated various constellation design criteria, such as turbo trellis-coded modulation [13], lattice modulation [14], joint optimization of MED and minimum product distance (MPD) [15], and channel-capacity optimization [16]. Meanwhile, the distribution of constellation points serves as an effective means to improve SCMA performance. For instance, Yu et al. have studied star-shaped quadrature amplitude modulation (QAM) and optimized the distance between inner and outer phase shift keying (PSK) rings [17]. Also, Zhou et al. have adopted a quadrature PSK (QPSK) constellation and applied rotation angles evenly spaced over 2π to generate different sub-constellations [18]. In [19], a golden angle modulation (GAM) has been introduced to construct the mother constellation. In addition to the code-domain CB designs, the joint design of power domain and code domain for CBs has been further explored for uplink SCMA [30] and downlink SCMA [31] to achieve greater robustness.

Another approach to mitigate inter-user interference is spreading, which is essentially a form of repetition coding. To be specific, each transmitted bit is mapped to a signature sequence that can ensure low inter-user correlation and enable separation at receiver via despreading [20], [21], [22], [23], [24]. Particularly, a repeat-accumulate (RA) code concatenated with block spreading for interleave division multiple access (IDMA) system has been first proposed to achieve good performance at low code rates [21]. To balance performance and transmission rate, both partially repeated spatially coupled RA code [22] and partially repeated low-density parity-check (LDPC) code have been proposed [23]. Additionally, a repetition-based automatic repeat request (ARQ) scheme has been investigated for the uplink SCMA system with LDPC codes [24].

Error-correction coding (ECC) is also an effective technology to enhance communication reliability, but traditional single-user codes may not be able to resist inter-user interference [11], [25], [26], [27]. To address this issue, significant research effort has been devoted to developing multi-user codes for NOMA scenarios. Specifically, some LDPC codes have been optimized via extrinsic information transfer (EXIT) analysis for the Gaussian multiple access channel (GMAC) [25]. Besides, the approximate message passing (AMP) algorithm has been employed to devise LDPC codes [28], [29]. In the above mentioned works, the LDPC codes offer excellent performance, but they are a type of un-structured ECCs having a quadratic encoding complexity with the codeword lengths [32]. For their hardware implementation, the codec consumes a lot of resources since the memory has to store a large parity-check matrix. As a structured variant, protograph-based LDPC (P-LDPC) codes are a more promising alternative, which can reduce the implementation complexity without sacrificing the performance advantage. Owing to their inherent benefits, several P-LDPC-code designs have also been studied for SCMA systems [11], [26], [27].

C. Motivation and Contributions

Although numerous studies have explored the SCMA designs, there remains room for performance enhancement. Specifically, (i) Existing CBs adopt an uniform UE-RE mapping that causes inadequate sparsity and excessive UEs sharing a single RE. Also, these CBs emphasize the mother constellation optimization instead of the superposition of multiple user signals in RE domain; (ii) Traditional repetition only treats the duplicate bits as redundant information and fails to fully explore the joint design of channel coding, resource mapping, and cross-slot data coupling; (iii) Most research works in ECC predominantly focus on either turbo or non-turbo decoding, thereby lacking a universal solution for different decoding paradigms. Motivated by the above three points, this paper concentrates on the design of non-uniform CB, segment-wise interleaved transmission (SWIT), and enhanced multi-user code (EMUC) for SCMA systems. The main contributions of this paper are summarized as follows.

1) We propose a two-step design method, which consists of low-density factor graph (LDFG) and energy-based constellation superposition (ECS) principles, to construct new non-uniform CBs. In particular, the LDFG and ECS principles are used to determine the non-uniform mapping relationship between UEs and REs as well as the superposition of multiple user symbols in RE domain, respectively.
2) To fully exploit the benefits of repetition-based transmission, we introduce a new SWIT scheme, which can systematically integrate repetition coding, resource mapping, and cross-slot data coupling (i.e., the coupling of data in adjacent time slots) to enhance SCMA transmission robustness.
3) As a further advancement, we conceive a decoding-threshold-guided multi-objective optimization (DT-MOO) method, and adopt multi-objective differential evolution (MODE) to design new protograph-based EMUCs. The designed EMUCs can achieve superior error-rate performance in SCMA systems with turbo and non-turbo decoding,[1] compared to traditional single-user and existing multi-user codes.
4) We carry out various simulations to demonstrate that the proposed non-uniform SCMA system with SWIT scheme and EMUCs is capable of achieving greater

[1]Turbo decoding refers to the iterative feedback of information between SCMA detector and P-LDPC decoder, while non-turbo decoding refers to non-iterative decoding, i.e., a single processing of SCMA detector and P-LDPC decoder.

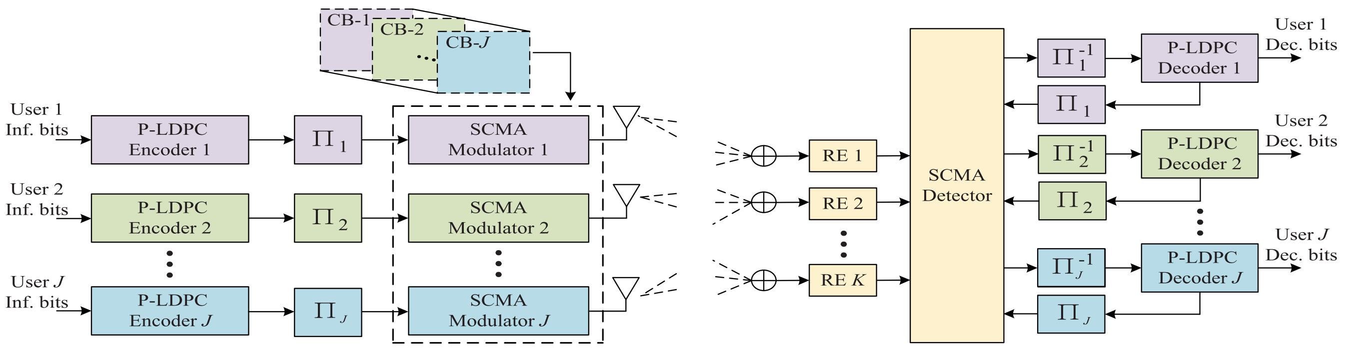


Fig. 1. Block diagram of a P-LDPC-coded SCMA system with J UEs and K REs over a Rayleigh fading channel.

robustness against inter-user interference than the state-of-the-art counterparts over Rayleigh fading channels.

Rayleigh fading effectively characterizes the signal amplitude fluctuations caused by multipath effect in wireless scenarios. The theoretical simulations conducted under time-varying Rayleigh fading channels then provide an accurate representation of real-world wireless propagation environments. Additionally, a majority of existing research works on SCMA consider the Rayleigh fading as the channel model [11], [15], [27].

The remainder of this paper is organized as follows. Sect. II introduces the system model of P-LDPC-coded SCMA framework. Sect. III details the design method of proposed non-uniform CBs. Sect. IV elaborates the operation mechanism of proposed SWIT scheme. Sect. V describes the proposed EMUCs and validates their superiorities from the perspective of convergence-performance analysis. Sect. VI provides various simulation results, and Sect. VII concludes the overall work.

II. P-LDPC-Coded SCMA System

Fig. 1 illustrates an uplink diagram of P-LDPC-coded SCMA system consisting of J independent UEs and K orthogonal REs. In particular, the system overload factor reflecting its overload capacity is $\lambda = J/K$ $(J > K)$. Also, owing to the inherent sparsity in SCMA, each UE occupies $O_j < K$ REs, while each RE is shared by $Q_k < J$ UEs. The mapping relationship between J UEs and K REs then can be captured by a factor graph matrix $\mathbf{F}_{K\times J}$. As seen from (1), the matrices $\mathbf{F}_{4\times 6}$ and $\mathbf{F}_{5\times 10}$ corresponding to the SCMA systems of sizes 4×6 and 5×10, respectively, are given as

$$\mathbf{F}_{4\times 6} = \begin{bmatrix} 1 & 1 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix},$$

$$\mathbf{F}_{5\times 10} = \begin{bmatrix} 1 & 1 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 0 & 1 & 0 & 1 & 1 \end{bmatrix}, \quad (1)$$

where $O_j = 2$, $Q_k = 3$ for $\mathbf{F}_{4\times 6}$, and $Q_k = 4$ for $\mathbf{F}_{5\times 10}$. In the above two matrices, the columns (resp. rows) correspond to the UEs (resp. REs), while the element '1' located in the j-th column and i-th row indicates that the UE j accesses RE i.

At the transmitting side, there are J independent information-bit sequences $\{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_J\}$ corresponding to J different UEs. Each sequence $\mathbf{b}_j$ $(j \in \{1, 2, \ldots, J\})$ will be encoded by its respective P-LDPC encoder into a coded-bit sequence $\mathbf{c}_j$, which is then permuted by an interleaver to an interleaved version $\hat{\mathbf{c}}_j$. Subsequently, every group of $m = \log_2 M$ coded bits extracted from $\hat{\mathbf{c}}_j$ is mapped into a K-dimensional complex-valued vector (i.e., symbol) $\mathbf{x}_j = [x_{1,j}, x_{2,j}, \ldots, x_{K,j}]^{\mathrm{T}}$ by SCMA modulator. Particularly, $\mathbf{x}_j$ is one of column vectors within the SCMA CB $\mathcal{X}_j$ with a cardinality $\left|\mathcal{X}^j\right| = M$. Consequently, J symbols from J distinct users are transmitted through a Rayleigh fading channel and superposed as

$$\mathbf{y} = \sum\nolimits_{j=1}^{J} \mathrm{diag}(\mathbf{h}_j)\mathbf{x}_j + \mathbf{n}, \quad (2)$$

where $\mathbf{y} = [y_1, y_2, \ldots, y_K]^{\mathrm{T}}$ and $\mathbf{h}_j = [h_{1,j}, h_{2,j}, \ldots, h_{K,j}]^{\mathrm{T}}$ denote the received signal vector and the channel gain vector for user j, respectively. Additionally, $\mathbf{n} \sim \mathcal{CN}(0, N_0)$ denotes a $K \times 1$ additive white Gaussian noise (AWGN) vector with zero mean and variance N_0 per complex dimension, where N_0 represents the noise power spectral density.

At the receiving side, to adequately mitigate the inter-user interference, multiplicative distortion, and additive noise in $\mathbf{y}$, a turbo decoding composed of both MPA detection [11] and belief propagation (BP) decoding [33], [34] is employed. In particular, the MPA detection is used to iteratively exchange soft information between user nodes (UNs) and resource nodes (RNs). During the first outer iteration,[2] the initial symbol probability of $\mathbf{x}_j$ is

$$g^0_{j\to k}(\mathbf{x}_j) = P(\mathbf{x}_j) = \frac{1}{M}, \qquad \forall j = 1, \ldots, J. \quad (3)$$

The MPA detection process is then shown as follows.

- **RN-to-UN Message Passing:** The message passed from the k-th RN to the j-th UN is expressed as

$$m^t_{k\to j}(\mathbf{x}_j) = \sum_{\mathbf{x}_i | i\in\mathcal{N}(k)\backslash j} \exp\Big\{ -\frac{1}{\sigma^2}\|y_k - \sum_{\substack{j=1,2,\ldots,J \\ x_{kj}\neq 0}} h_{kj}x_{kj}\|^2 \Big\} \prod_{i\in\mathcal{N}(k)\backslash j} g^{t-1}_{i\to k}(\mathbf{x}_i), \quad (4)$$

[2] The outer iteration (i.e., T_{out}) denotes information exchange between SCMA detector and BP decoder, while both MPA and BP iterations (i.e., T_{MPA} and T_{BP}) refer to inner iterations within SCMA detector and BP decoder, respectively.

where $g^{t-1}_{i\to k}(\mathbf{x}_i)$ denotes the message passed from the i-th $(i \neq j)$ UN to the k-th RN, and $\mathcal{N}(k) \setminus j$ denotes the set of users sharing RE k except UE j.

- **UN-to-RN Message Passing:** The message passed from the j-th UN to the k-th RN is expressed as

$$\overline{g^{t}_{j\to k}}(\mathbf{x}_j) = \frac{1}{M} \prod_{i\in\mathcal{R}(j)\setminus k} m^{t-1}_{i\to j}(\mathbf{x}_j), \tag{5}$$

where $\mathcal{R}(j) \setminus k$ denotes the set of resources occupied by UE j except RE k.

- **Normalization of Symbol LLR:** The normalized message passed from the j-th UN to the k-th RN is expressed as

$$g^{t}_{j\to k}(\mathbf{x}_j) = \frac{\overline{g^{t}_{j\to k}}(\mathbf{x}_j)}{\sum_{j=1}^{J} \overline{g^{t}_{j\to k}}(\mathbf{x}_j)}. \tag{6}$$

The MPA process will terminates once the maximum number of MPA iterations T_{MPA} is reached. Subsequently, the *a-posteriori* probabilities of $\mathbf{x}_j$ can be calculated as

$$\mathbf{Pr}(\mathbf{x}_j) = \frac{1}{M} \prod_{k\in\mathcal{R}(j)} m^{T_{\text{MPA}}}_{k\to j}(\mathbf{x}_j). \tag{7}$$

After that, $\mathbf{Pr}(\mathbf{x}_j)$ is converted into the bit-level LLRs of the corresponding $\log_2 M$ coded bits, as

$$L_{a,j}(b_i) = \log \frac{\sum_{\{\mathbf{x}_j\in\mathcal{X}|b_i=0\}} \mathbf{Pr}(\mathbf{x}_j)}{\sum_{\{\mathbf{x}_j\in\mathcal{X}|b_i=1\}} \mathbf{Pr}(\mathbf{x}_j)},\ \forall i = 1,\ldots,\log_2\ M. \tag{8}$$

Consequently, $L_{a,j}(b_i)$ is passed to its corresponding P-LDPC decoder, and used as the *a-priori* LLR for the P-LDPC decoding, which is implemented by the BP algorithm. In the subsequent outer iterations, $P(\mathbf{x}_j)$ is updated based on the *a-posteriori* LLR provided by the P-LDPC decoder.

III. Proposed Non-Uniform Codebooks

The codebook (CB) determines the sparse mapping of user data to multiple shared REs, serving as a fundamental mechanism to enable SCMA. However, owing to the superposition of the transmitted symbols from different UEs on a certain RE, this inevitably leads to the inter-user interference. For the existing CB designs, there is still room for performance improvement.

- As observed from (1), the existing CBs possess an uniform feature in the sparse mapping between the UEs and the REs. Specifically, for $\mathbf{F}_{4\times6}$ (resp. $\mathbf{F}_{5\times10}$), one can easily notice that the number of REs occupied by each UE is identical (i.e., $O_j = 2$), while every RE serves the same number of UEs $Q_k = 3$ (resp. $Q_k = 4$). It is evident that the sparsity of the factor graph matrix can be further enhanced, thereby reducing the number of UEs sharing the same RE. This helps to mitigate inter-user interference, which is the most critical factor deteriorating SCMA reliability.
- Most CB designs rely on optimizing constellation operators (e.g., phase rotation, permutation, and complex conjugation) for the mother constellation. Obviously, these approaches primarily focus on the distribution of single-user signals in modulation space, while ignoring the structural properties of multi-user symbol superposition on the same RE. In contrast, investigating the symbol superposition from the perspective of REs may capture the interference feature among different UEs sharing the same RE and provide a new direction for CB optimization to mitigate inter-user interference.

Inspired by the above discussion, we propose a non-uniform CB design method, which consists of LDFG and ECS optimization principles. More details are shown below.

1) **LDFG Optimization:** For an uplink SCMA system comprising $J \in \mathbb{N}$ UEs $\mathcal{U} = \{u_1, u_2, \ldots, u_J\}$ and $K \in \mathbb{N}$ orthogonal REs $\mathcal{R} = \{r_1, r_2, \ldots, r_K\}$, the overload factor is then $\lambda \triangleq \frac{J}{K}$.[3] An initial UE-to-RE mapping represented by an all-zero factor graph matrix $\mathbf{F} \in \{0\}^{K\times J}$ can be assumed first. Subsequently, $\mathbf{F}$ is updated to satisfy a hybrid sparsity constraint as

$$\begin{cases} \|\mathbf{f}_j\|_0 = 1, & j \leq K, \quad \text{(orthogonal)} \\ \|\mathbf{f}_j\|_0 = d_v \geq 2, & j > K, \quad \text{(overloaded)} \end{cases} \tag{9}$$

where $\mathbf{f}_j$ and $\|\cdot\|_0$ denote the j-th column of $\mathbf{F}$ and L_0-norm of a vector, respectively. Obviously, the submatrix $\mathbf{F}_{:,1:K}$ is equal to an identity matrix $\mathbf{I}_{K\times K}$, satisfying the condition of $\sum_{k=1}^{K} f_{k,j} = 1$ for $j = 1, 2, \ldots, K$. This help ensure the orthogonality for the first K UEs. Additionally, for the remaining $J' = (\lambda - 1)K$ UEs, they access d_v REs according to the following assignment condition

$$\|\mathbf{f}_j\|_0 = d_v, \quad d_v K \geq j > (d_v - 1)K. \tag{10}$$

Also, to reduce the number of cycles (i.e., the number of overlapping non-zero positions between any two column vectors) during the factor graph expansion, the resultant $\mathbf{F}$ must satisfy the following cycle constraint

$$\text{Minimize: } |\text{supp}(\mathbf{f}_i) \cap \text{supp}(\mathbf{f}_j)|,\ \text{(preferably achieve 0)}, \tag{11}$$

where $\mathbf{f}_z = [f_{1,z}, f_{2,z}, \ldots, f_{K,z}]^{\mathrm{T}}$ denotes the z-th column of $\mathbf{F}$, $\text{supp}(\mathbf{f}_z) = \{k|f_{k,z} \neq 0\}$ denotes the index set of non-zero entries in $\mathbf{f}_z$ (i.e., the indices of REs occupied by UE u_z), $z \in \{i, j\}$, and $i \neq j$. After the above processing, two specific matrices $\mathbf{F}'_{4\times6}$ and $\mathbf{F}'_{5\times10}$ corresponding to the non-uniform CBs, respectively, are given as

$$\mathbf{F}'_{4\times6} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 \\ 0 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 0 & 1 \end{bmatrix},$$
$$\mathbf{F}'_{5\times10} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 \\ 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 1 \end{bmatrix}. \tag{12}$$

The proposed factor graph design aims to (i) reduce the number of REs occupied by each user, (ii) prevent a

[3]Similar to the previously-published CB works [7], [10], [11], [15], [27], [35], [36], this paper only focuses on the case of $1 < \lambda \leq 2$ for simplicity.

complete overlap of occupied REs among users, and (iii) minimize the number of 4-cycles, thereby effectively avoiding a severe signal collision and mitigating the error propagation phenomenon.

2) **ECS Optimization:** After the above design, a low-density non-uniform factor graph $\mathbf{F}$ depicting the sparse mapping between UEs and REs is determined. The UE energy is then allocated to the active edges within $\mathbf{F}$ (i.e., $E_{k,j} \neq 0$ if $f_{k,j} = 1$ for $k = 1, 2, \ldots, K$ and $j = 1, 2, \ldots, J$). Specifically, for each RE r_k accessed by $Q_k = \sum_{j=1}^{J} f_{k,j}$ UEs, the corresponding Q_k-level energy vector is defined as

$$\mathbf{E}_k = \left[E_k^1, E_k^2, \ldots, E_k^{Q_k}\right] \in \mathbb{R}_+^{Q_k}, \tag{13}$$

subject to the following constraint

$$\frac{E_k^1}{Q_k} = \frac{E_k^2}{Q_k - 1} = \cdots = \frac{E_k^{Q_k}}{1}, \tag{14}$$

where E_k^1 is assigned to the non-zero position $f_{k,k}$ within $\mathbf{f}_j \in \mathbf{F}_{:,1:K}$, and the remaining entries $E_k^2, E_k^3, \ldots, E_k^{Q_k}$ are assigned to $Q_k - 1$ non-zero positions within each row vector of $\mathbf{F}_{:,K+1:J}$. Also, the resultant energy matrix $\mathbf{E} \in \mathbb{R}_+^{K \times J}$ must satisfy the user-energy normalization constraint

$$\sum\nolimits_{k=1}^{K} E_{k,j} = P_j, \quad \forall j = 1, 2, \ldots, J, \tag{15}$$

where P_j is the pre-set energy of the j-th user. Given a determined $\mathbf{E} = [E_{k,j}] \in \mathbb{R}_+^{K \times J}$, the high-order modulation scheme can be implemented for all J UEs. Specifically, for the first K UEs corresponding to the first K columns of $\mathbf{F}$, each of which accesses only a single RE. The normalized energy of the modulation symbol is equal to

$$\sum\nolimits_{i=1}^{M} |\mathbf{s}_{(j,j)}^i|^2 / M = P_j, \quad j = 1, 2, \ldots, K, \tag{16}$$

where $\mathbf{s}_{(j,j)}^i$ is the i-th complex-valued modulation symbol located at the j-th row and the j-th column of $\mathbf{F}$. For the remaining $J - K$ UEs, each of which accesses d_v REs, and then the index set of active REs for the j-th ($j = K + 1, \ldots, J$) UE can be defined as

$$\mathcal{K}_j = \{k \in \{1, 2, \ldots, K\} \mid E_{k,j} \neq 0\}, \quad |\mathcal{K}_j| = d_v. \tag{17}$$

Accordingly, for the j-th UE, the normalized energy of the complex-valued modulation symbol located at the k-th row and the j-th column of $\mathbf{F}$ is set to

$$\sum\nolimits_{i=1}^{M} |\mathbf{s}_{(k,j)}^i|^2 / M = E_{k,j}, \quad k \in \mathcal{K}_j. \tag{18}$$

The above energy allocation ensures that, even when multiple users' signals overlap on the same RE, the receiver can still effectively recover the original signals based on their energy differences. On the other hand, the resultant non-uniform CB causes different performance across users, but the least-reliable one can outperform that employed with those traditional CBs (see Fig. 7 in Sect. VI-A).

The proposed LDFG and ECS principles does not involve the modulation order. Therefore, similar to most CB-related research studies [11], [18], the QPSK modulation is assumed in this work. Due to the user-energy difference among the Q_k UEs sharing the same RE, a 4^{Q_k}-ary constellation is constructed by superposing Q_k different QPSK constellations, while the 4^{Q_k} constellation points avoid overlap as much as possible.

To elaborate the above design, Fig. 2 presents a constellation superposition example for 4×6 SCMA system with a non-uniform CB. Furthermore, the details of two different non-uniform CBs for 4×6 and 5×10 SCMA systems are respectively provided in (19) and (20). Here, $\mathcal{X}_{j,\{k_1\}}$ (resp. $\mathcal{X}_{j,\{k_1,k_2\}}$) denotes that the j-th UE occupies the k_1-th RE (resp. both k_1-th and k_2-th REs). Additionally, the entries $\mathcal{E}_1$, $\mathcal{E}_2$, $\mathcal{E}_3$, $\mathcal{E}_4$, and $\mathcal{E}_5$ are respectively set to 0.7071, 0.5000, 1.0000, 0.8165, and 0.5774.

$$\begin{aligned}
\mathcal{X}_{1,\{1\}} &= \begin{bmatrix} \mathcal{E}_1 + \mathcal{E}_1 i, & \mathcal{E}_1 - \mathcal{E}_1 i, & -\mathcal{E}_1 + \mathcal{E}_1 i, & -\mathcal{E}_1 - \mathcal{E}_1 i \end{bmatrix}, \\
\mathcal{X}_{2,\{2\}} &= \begin{bmatrix} \mathcal{E}_1 + \mathcal{E}_1 i, & \mathcal{E}_1 - \mathcal{E}_1 i, & -\mathcal{E}_1 + \mathcal{E}_1 i, & -\mathcal{E}_1 - \mathcal{E}_1 i \end{bmatrix}, \\
\mathcal{X}_{3,\{3\}} &= \begin{bmatrix} \mathcal{E}_1 + \mathcal{E}_1 i, & \mathcal{E}_1 - \mathcal{E}_1 i, & -\mathcal{E}_1 + \mathcal{E}_1 i, & -\mathcal{E}_1 - \mathcal{E}_1 i \end{bmatrix}, \\
\mathcal{X}_{4,\{4\}} &= \begin{bmatrix} \mathcal{E}_1 + \mathcal{E}_1 i, & \mathcal{E}_1 - \mathcal{E}_1 i, & -\mathcal{E}_1 + \mathcal{E}_1 i, & -\mathcal{E}_1 - \mathcal{E}_1 i \end{bmatrix}, \\
\mathcal{X}_{5,\{1,2\}} &= \begin{bmatrix} \mathcal{E}_2 + \mathcal{E}_2 i, & \mathcal{E}_2 - \mathcal{E}_2 i, & -\mathcal{E}_2 + \mathcal{E}_2 i, & -\mathcal{E}_2 - \mathcal{E}_2 i \\ -\mathcal{E}_2 - \mathcal{E}_2 i, & -\mathcal{E}_2 + \mathcal{E}_2 i, & \mathcal{E}_2 - \mathcal{E}_2 i, & \mathcal{E}_2 + \mathcal{E}_2 i \end{bmatrix}, \\
\mathcal{X}_{6,\{3,4\}} &= \begin{bmatrix} \mathcal{E}_2 + \mathcal{E}_2 i, & \mathcal{E}_2 - \mathcal{E}_2 i, & -\mathcal{E}_2 + \mathcal{E}_2 i, & -\mathcal{E}_2 - \mathcal{E}_2 i \\ -\mathcal{E}_2 - \mathcal{E}_2 i, & -\mathcal{E}_2 + \mathcal{E}_2 i, & \mathcal{E}_2 - \mathcal{E}_2 i, & \mathcal{E}_2 + \mathcal{E}_2 i \end{bmatrix}.
\end{aligned} \tag{19}$$

$$\begin{aligned}
\mathcal{X}_{1,\{1\}} &= \begin{bmatrix} \mathcal{E}_3 + \mathcal{E}_3 i, & \mathcal{E}_3 - \mathcal{E}_3 i, & -\mathcal{E}_3 + \mathcal{E}_3 i, & -\mathcal{E}_3 - \mathcal{E}_3 i \end{bmatrix}, \\
\mathcal{X}_{2,\{2\}} &= \begin{bmatrix} \mathcal{E}_3 + \mathcal{E}_3 i, & \mathcal{E}_3 - \mathcal{E}_3 i, & -\mathcal{E}_3 + \mathcal{E}_3 i, & -\mathcal{E}_3 - \mathcal{E}_3 i \end{bmatrix}, \\
\mathcal{X}_{3,\{3\}} &= \begin{bmatrix} \mathcal{E}_3 + \mathcal{E}_3 i, & \mathcal{E}_3 - \mathcal{E}_3 i, & -\mathcal{E}_3 + \mathcal{E}_3 i, & -\mathcal{E}_3 - \mathcal{E}_3 i \end{bmatrix}, \\
\mathcal{X}_{4,\{4\}} &= \begin{bmatrix} \mathcal{E}_3 + \mathcal{E}_3 i, & \mathcal{E}_3 - \mathcal{E}_3 i, & -\mathcal{E}_3 + \mathcal{E}_3 i, & -\mathcal{E}_3 - \mathcal{E}_3 i \end{bmatrix}, \\
\mathcal{X}_{5,\{5\}} &= \begin{bmatrix} \mathcal{E}_3 + \mathcal{E}_3 i, & \mathcal{E}_3 - \mathcal{E}_3 i, & -\mathcal{E}_3 + \mathcal{E}_3 i, & -\mathcal{E}_3 - \mathcal{E}_3 i \end{bmatrix}, \\
\mathcal{X}_{6,\{1,2\}} &= \begin{bmatrix} \mathcal{E}_4 + \mathcal{E}_4 i, & \mathcal{E}_4 - \mathcal{E}_4 i, & -\mathcal{E}_4 + \mathcal{E}_4 i, & -\mathcal{E}_4 - \mathcal{E}_4 i \\ -\mathcal{E}_5 - \mathcal{E}_5 i, & -\mathcal{E}_5 + \mathcal{E}_5 i, & \mathcal{E}_5 - \mathcal{E}_5 i, & \mathcal{E}_5 + \mathcal{E}_5 i \end{bmatrix}, \\
\mathcal{X}_{7,\{2,3\}} &= \begin{bmatrix} \mathcal{E}_4 + \mathcal{E}_4 i, & \mathcal{E}_4 - \mathcal{E}_4 i, & -\mathcal{E}_4 + \mathcal{E}_4 i, & -\mathcal{E}_4 - \mathcal{E}_4 i \\ -\mathcal{E}_5 - \mathcal{E}_5 i, & -\mathcal{E}_5 + \mathcal{E}_5 i, & \mathcal{E}_5 - \mathcal{E}_5 i, & \mathcal{E}_5 + \mathcal{E}_5 i \end{bmatrix}, \\
\mathcal{X}_{8,\{3,4\}} &= \begin{bmatrix} \mathcal{E}_4 + \mathcal{E}_4 i, & \mathcal{E}_4 - \mathcal{E}_4 i, & -\mathcal{E}_4 + \mathcal{E}_4 i, & -\mathcal{E}_4 - \mathcal{E}_4 i \\ -\mathcal{E}_5 - \mathcal{E}_5 i, & -\mathcal{E}_5 + \mathcal{E}_5 i, & \mathcal{E}_5 - \mathcal{E}_5 i, & \mathcal{E}_5 + \mathcal{E}_5 i \end{bmatrix}, \\
\mathcal{X}_{9,\{4,5\}} &= \begin{bmatrix} \mathcal{E}_4 + \mathcal{E}_4 i, & \mathcal{E}_4 - \mathcal{E}_4 i, & -\mathcal{E}_4 + \mathcal{E}_4 i, & -\mathcal{E}_4 - \mathcal{E}_4 i \\ -\mathcal{E}_5 - \mathcal{E}_5 i, & -\mathcal{E}_5 + \mathcal{E}_5 i, & \mathcal{E}_5 - \mathcal{E}_5 i, & \mathcal{E}_5 + \mathcal{E}_5 i \end{bmatrix}, \\
\mathcal{X}_{10,\{1,5\}} &= \begin{bmatrix} \mathcal{E}_4 + \mathcal{E}_4 i, & \mathcal{E}_4 - \mathcal{E}_4 i, & -\mathcal{E}_4 + \mathcal{E}_4 i, & -\mathcal{E}_4 - \mathcal{E}_4 i \\ -\mathcal{E}_5 - \mathcal{E}_5 i, & -\mathcal{E}_5 + \mathcal{E}_5 i, & \mathcal{E}_5 - \mathcal{E}_5 i, & \mathcal{E}_5 + \mathcal{E}_5 i \end{bmatrix}.
\end{aligned} \tag{20}$$

IV. Proposed Segment-Wise Interleaved Transmission Scheme

To mitigate inter-user interference in SCMA systems, repetition coding is an effective method, which operates by repeating coded bits at the transmitting side and performing soft information merging at the receiving side. However, the traditional repetition coding only treats the duplicate bits as redundant information, leading to a limited performance improvement. To be specific,

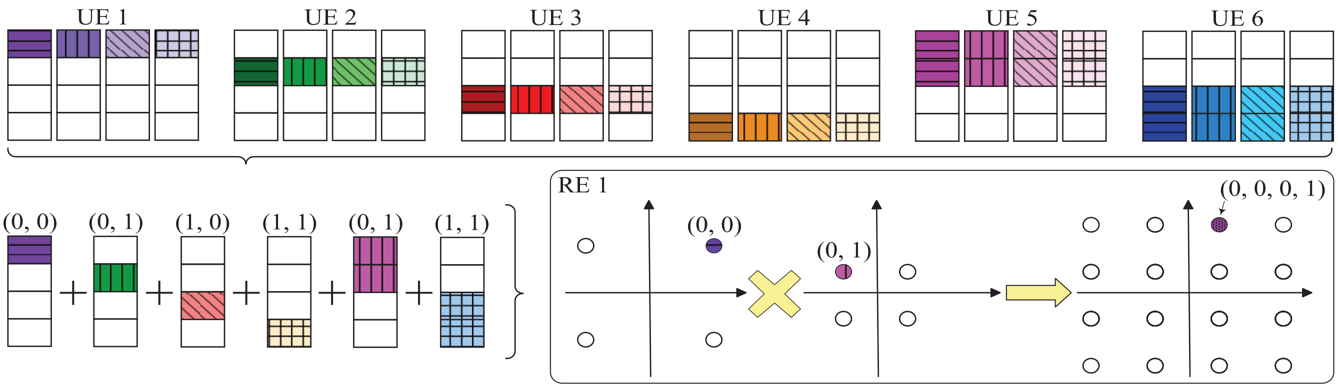


Fig. 2. Mapping relationship between users and resources in the proposed 4×6 non-uniform CB and the corresponding signal superposition on RE 1.

- The traditional repetition does not incorporate coding and modulation into a joint design framework, making repeated information difficult to provide a large decoding gain;
- Additionally, the traditional repetition design is confined to a single data slot (i.e., lacks cross-slot data coupling), thereby failing to achieve a cumulative information gain from the time domain.

To further explore the potential of repetition-based transmission, we propose a structured transmission method, called *segment-wise interleaved transmission (SWIT)*. The proposed SWIT aims to introduce a systematic optimization among repetition coding, resource mapping, and cross-slot data coupling, while its design principle is illustrated as follows.

A. Transmitting-Side SWIT Mechanism

The SWIT transmitting mechanism consists of two interleaved procedures: (i) interleaved block-delay mapping for the original data and the 1-st duplicate, and (ii) interleaved Markovian superposition for the remaining duplicates.

1) **Parameter Initialization:** Three key parameters are defined as
   - the index of slot $t \in \mathbb{Z}^+$.
   - the number of repetitions per slot $q \in \mathbb{Z}^+$;
   - the number of coded blocks transmitted per slot before repetition $B \in \mathbb{Z}^+$;

   Then, each slot contains a total of $(q+1)B$ coded blocks.
2) **Segment-Wise Data Partitioning:** Let the original B coded blocks transmitted at slot t be $\{\mathbf{c}_{(t,1)}, \mathbf{c}_{(t,2)}, \ldots, \mathbf{c}_{(t,B)}\}$, where $\mathbf{c}_{(t,b)} \in \{0,1\}^n$ is the b-th coded block of length n. After q repetitions, $(q+1)B$ blocks are divided into three parts:
   Part 1: original coded blocks $\{\mathbf{c}_{(t,1)}, \ldots, \mathbf{c}_{(t,B)}\}$;
   Part 2: 1-st duplicate $\{\mathbf{c}_{(t,1,1)}, \ldots, \mathbf{c}_{(t,1,B)}\}$;
   Part 3: remaining $q-1$ duplicates $\{\mathbf{c}_{(t,r,1)}, \ldots, \mathbf{c}_{(t,r,B)}\}_{r=2}^{q}$.
3) **Interleaved Block-Delay Mapping:** To facilitate a joint design on channel coding and resource mapping, the coded blocks in both Parts 1 & 2 are grouped together. Specifically, given an all-zero block $\mathbf{0} \in \{0,1\}^n$ and a group index number $g \in \{1, 2, \ldots, B+1\}$, the grouping rule is depicted as

$$\begin{cases} \mathcal{G}_1 : & \{\mathbf{c}_{(t,1)}, \mathbf{0}\}, \\ \mathcal{G}_g \ (2 \leq g = b \leq B) : & \{\mathbf{c}_{(t,b)}, \mathbf{c}_{(t,1,b-1)}\}, \\ \mathcal{G}_{B+1} : & \{\mathbf{c}_{(t,1,B)}\}. \end{cases} \tag{21}$$

   Subsequently, for the group $\mathcal{G}_g$ $(g \leq B)$, we perform a bit-wise interleaving to constitute the modulation symbol, as
   - Sequentially extract one bit from the first coded block and the second coded block to form a QPSK symbol.

   For the last group $\mathcal{G}_{B+1}$, two bits are extracted sequentially from a single coded block to form a QPSK symbol. To make this mapping procedure clearer, Fig. 3 depicts the details.

   Property 1: When the coded block $\mathbf{c}_{(t,b)} \in \mathcal{G}_g$ is decoded, its soft information (i.e., LLRs) can assist the detection of $\mathbf{c}_{(t,b+1)} \in \mathcal{G}_{g+1}$, thereby improving interference cancelation for $\mathbf{c}_{(t,b+1)}$, where $b = 1, 2, \ldots, B-1$.
4) **Interleaved Markovian Superposition:** To reduce the number of error bits at a transmission slot, we introduce Markovian superposition to establish a structured dependency relationship among different coded blocks over successive slots. Such a superposition mechanism enables a soft information exchange across different transmission data frames. Here, we define the remaining $(q-1)B$ coded blocks (i.e., Part 3) at slot t as $\{\mathbf{c}_{(t,r,1)}, \mathbf{c}_{(t,r,2)}, \ldots, \mathbf{c}_{(t,r,B)}\}_{r=2}^{q} = \{\mathcal{C}_{(t,2)}, \mathcal{C}_{(t,3)}, \ldots, \mathcal{C}_{(t,q)}\}$, where $\{\mathbf{c}_{(t,r,1)}, \mathbf{c}_{(t,r,2)}, \ldots, \mathbf{c}_{(t,r,B)}\} = \mathcal{C}_{(t,r)}$. Moreover, we define an interleaved version of Part 3 at the previous slot $t-1$ as $\{\hat{\mathcal{C}}_{(t-1,2)}, \hat{\mathcal{C}}_{(t-1,3)}, \ldots, \hat{\mathcal{C}}_{(t-1,q)}\}$, where $\hat{\mathcal{C}}_{(t-1,r)}$ for $r = 2, 3, \ldots, q$ is formulated by adopting a cyclic-bit-interleaved strategy across B coded blocks, as

$$\begin{aligned} &\hat{\mathcal{C}}_{(t-1,r)} \\ &= \{\mathbf{c}_{(t-1,r,1)}(i), \mathbf{c}_{(t-1,r,2)}(i), \ldots, \mathbf{c}_{(t-1,r,B)}(i)\}_{i=1}^{n}, \\ &= \{\hat{\mathbf{c}}_{(t-1,r,1)}, \hat{\mathbf{c}}_{(t-1,r,2)}, \ldots, \hat{\mathbf{c}}_{(t-1,r,B)}\}. \end{aligned} \tag{22}$$

   Based on the bit-wise XOR between $\{\mathcal{C}_{(t,2)}, \mathcal{C}_{(t,3)}, \ldots, \mathcal{C}_{(t,q)}\}$ and $\{\hat{\mathcal{C}}_{(t-1,2)}, \hat{\mathcal{C}}_{(t-1,3)}, \ldots, \hat{\mathcal{C}}_{(t-1,q)}\}$, the superposition blocks $\{\mathcal{S}_{(t,2)}, \mathcal{S}_{(t,3)}, \ldots, \mathcal{S}_{(t,q)}\}$

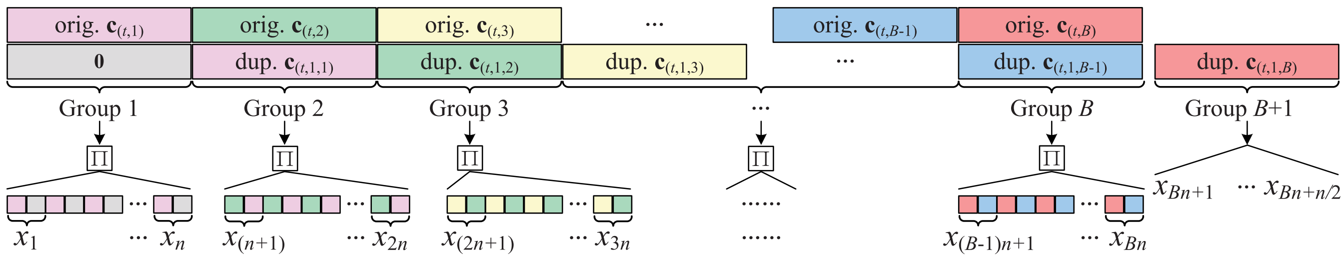


Fig. 3. Interleaved block-delay mapping (i.e., the first step) of the proposed SWIT scheme.

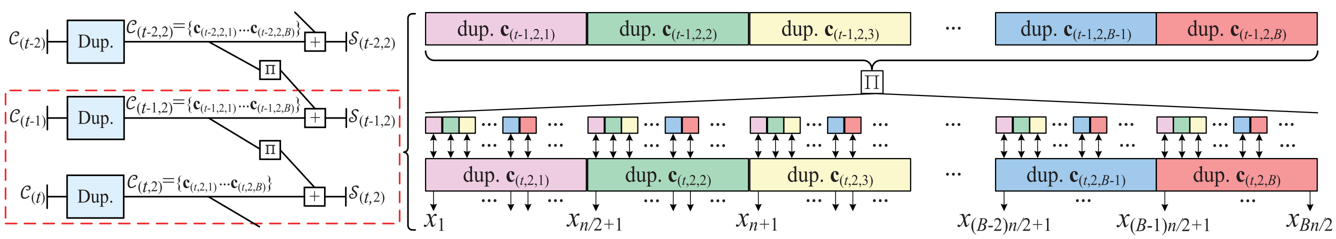


Fig. 4. Interleaved Markovian superposition coding (i.e., the second step) of the proposed SWIT scheme. In particular, the repetition parameter is $q = 2$.

are obtained, where $\mathcal{S}_{(t,r)} = \mathcal{C}_{(t,r)} \oplus \hat{\mathcal{C}}_{(t-1,r)}$ $(r = 2, 3, \ldots, q)$ and more details are shown as

$$\begin{cases} \mathcal{S}_{(t,r)} = \{\mathbf{s}_{(t,r,1)}, \mathbf{s}_{(t,r,2)}, \ldots, \mathbf{s}_{(t,r,B)}\} \\ \mathcal{C}_{(t,r)} = \{\mathbf{c}_{(t,r,1)}, \mathbf{c}_{(t,r,2)}, \ldots, \mathbf{c}_{(t,r,B)}\} \\ \hat{\mathcal{C}}_{(t-1,r)} = \{\hat{\mathbf{c}}_{(t-1,r,1)}, \hat{\mathbf{c}}_{(t-1,r,2)}, \ldots, \hat{\mathbf{c}}_{(t-1,r,B)}\} \end{cases}$$
$$\mathbf{s}_{(t,r,b)} = \{\mathbf{s}_{(t,r,b)}(1), \mathbf{s}_{(t,r,b)}(2), \ldots, \mathbf{s}_{(t,r,b)}(n)\},$$
$$\mathbf{s}_{(t,r,b)}(i) = \mathbf{c}_{(t,r,b)}(i) \oplus \hat{\mathbf{c}}_{(t-1,r,b)}(i). \tag{23}$$

Property 2: As seen from Fig. 4, the Markovian superposition induces a time-domain Markov chain between adjacent slots. The decoding of coded block thus depends on both current and next observations. Specifically, when $\mathbf{c}_{(t,b)}$ cannot be correctly decoded at slot t, an iterative decoding between two adjacent slots t and $t+1$ may enable its successful recovery.

Remark 1: Assuming each coded block carries k information bits within n coded bits, the spectral efficiency of traditional repetition transmission is

$$R_{\mathrm{RE}} = k/(n(q+1)), \tag{24}$$

while that of the proposed SWIT is

$$R_{\mathrm{SWIT}} = Bk/(Bn(q+1)+n). \tag{25}$$

By comparing (24) with (25), one can notice that the rate loss of the proposed SWIT is only $R_{\mathrm{RE}}/(Bq+B+1)$, and it can be reduced as the numbers of transmitted coded blocks B and repetitions q increase. Importantly, a large value of B enhances the discreteness of distribution of unreliable coded bits at the previous slot t when performing the interleaved Markovian superposition at the current slot $t+1$, and thus the intrinsic error propagation incurred by Markovian superposition can be effectively eliminated.

Note that spatially coupled LDPC (SC-LDPC) coding is a type of error-correction codes [37], and processes the information bits. By integrating Markovian superposition technique into SC-LDPC coding, one can construct superposition-based SC-LDPC coding [38], which is a class of block Markov superposition transmission (BMST) coding. Similarly, the proposed interleaved Markovian superposition coding can be also regarded as the BMST coding. However, the proposed design is introduced specifically for SCMA system with repetition coding. The is because the repetition coding treats duplicate bits merely as redundant information and leaves significant performance gains to be achieved. Hence, the proposed design enables further performance improvements and marks the first application of BMST coding in SCMA communications.

Based on the proposed interleaved Markovian superposition coding, each data frame conveys information from both previous and current slots. In other words, a single data frame is delivered twice without requiring the retransmission mechanism, thereby significantly contributing to the energy-efficient transmission.

B. Receiving-Side SWIT Mechanism

The SWIT receiving mechanism is presented in *Algorithm 1* composed of three data-processing stages: (i) A-priori information calculation for superposition data at current slot; (ii) Sequential decoding on group data at current slot; and (iii) Superposition decoding on superposition data across two adjacent slots.

1) **A-Priori Information Calculation:** At slot t, the received superposition data is first detected by MPA to obtain the corresponding *a-priori* LLRs $\{\mathcal{L}^{\mathrm{h,a}}_{(t,r)}\}_{r=2}^{q}$, where $\mathcal{L}^{\mathrm{h,a}}_{(t,r)} = \{\mathbf{l}^{\mathrm{h,a}}_{(t,r,b)}\}_{b=1}^{B}$. An XOR is then used to calculate the *a-priori* LLRs of Part-3 data $\{\mathcal{L}^{\mathrm{s,a}}_{(t,r)}\}_{r=2}^{q}$, where $\mathcal{L}^{\mathrm{s,a}}_{(t,r)} = \{\mathbf{l}^{\mathrm{s,a}}_{(t,r,b)}\}_{b=1}^{B}$. Subsequently, $\{\mathcal{L}^{\mathrm{s,a}}_{(t,r)}\}_{r=2}^{q}$ is combined to $\mathcal{L}^{\mathrm{s,a}}_{(t)} = \{\mathbf{l}^{\mathrm{s,a}}_{(t,b)}\}_{b=1}^{B}$, which is used to assist the following sequential decoding on group data. Here, the XOR processing procedure is omitted for simplicity, but its details can be found in (27).
2) **Group-Wise Sequential Decoding:** The received group data has $B+1$ groups $\{\mathcal{G}_1, \mathcal{G}_2, \ldots, \mathcal{G}_{B+1}\}$. Each

of the first B groups includes two coded blocks $\{\mathbf{c}_{(t,b)}, \mathbf{c}_{(t,1,b-1)}\}$ for $b = 1, 2, \ldots, B$, where $\mathbf{c}_{(t,1,0)}$ is an all-zero coded block. The last (i.e., $(B+1)$-th) group is a single coded block $\mathbf{c}_{(t,1,B)}$.

For $\mathcal{G}_1$, an MPA is first performed with known *a-posteriori* LLRs $\mathbf{l}^{\mathrm{p}}_{(t,1,0)}$ of $\mathbf{c}_{(t,1,0)}$ to obtain the *a-priori* LLRs $\mathbf{l}^{\mathrm{a}}_{(t,1)}$. Then, based on $\mathbf{l}^{\mathrm{a}}_{(t,1)}$ and $\mathbf{l}^{\mathrm{s,a}}_{(t,1)}$, a BP decoding is performed on $\mathbf{c}_{(t,1)}$ to obtain its *a-posteriori* LLRs $\mathbf{l}^{\mathrm{p}}_{(t,1)}$. Similarly, since $\mathbf{c}_{(t,1)}$ is the same as $\mathbf{c}_{(t,1,1)}$, $\mathbf{l}^{\mathrm{p}}_{(t,1)}$ is used to assist the detection of $\mathcal{G}_2$ to obtain the *a-priori* LLRs $\mathbf{l}^{\mathrm{a}}_{(t,2)}$. Specifically, two different decoding cases are considered, as follows

(i) If $\mathbf{c}_{(t,1)}$ is correctly decoded in the previous decoding, the BP algorithm is then executed on $\mathbf{c}_{(t,2)}$ based on the *a-priori* LLRs $\mathbf{l}^{\mathrm{a}}_{(t,2)}$ and $\mathbf{l}^{\mathrm{s,a}}_{(t,2)}$ to obtain its *a-posteriori* LLRs $\mathbf{l}^{\mathrm{p}}_{(t,2)}$.

(ii) Otherwise, based on the *a-priori* LLRs $\mathbf{l}^{\mathrm{a}}_{(t,1)}$, $\mathbf{l}^{\mathrm{s,a}}_{(t,1)}$ and $\mathbf{l}^{\mathrm{a}}_{(t,1,1)}$ (obtained from the detection of $\mathcal{G}_2$ having $\mathbf{c}_{(t,1,1)}$), a BP decoding can be implemented on $\mathbf{c}_{(t,1)}$ again. In particular,

- Decoding Success: Based on the improved $\mathbf{l}^{\mathrm{p}}_{(t,1)}$, one can perform MPA on $\mathcal{G}_2$ again to obtain improved *a-priori* LLRs $\mathbf{l}^{\mathrm{a}}_{(t,2)}$ of $\mathbf{c}_{(t,2)}$. Then, the BP decoding is executed on $\mathbf{c}_{(t,2)}$ to obtain its *a-posteriori* LLRs $\mathbf{l}^{\mathrm{p}}_{(t,2)}$.
- Decoding Failure: The BP decoding is directly performed on $\mathbf{c}_{(t,2)}$ to obtain its *a-posteriori* LLRs $\mathbf{l}^{\mathrm{p}}_{(t,2)}$.

Based on the *a-posteriori* LLRs $\mathbf{l}^{\mathrm{p}}_{(t,2)}$, such a group-wise sequential decoding procedure can be applied to $\mathcal{G}_3$. The similar operation is then performed sequentially on the remaining groups (i.e., $\mathcal{G}_4, \mathcal{G}_5, \ldots, \mathcal{G}_{B+1}$).

3) **Slot-Wise Superposition Decoding:** When the decoding of data transmitted at slot $t-1$ fails, its superposition decoding is triggered at slot t. We define the maximum number of iterations in superposition decoding as T_{SD}, and elaborate the superposition decoding as follows.

(i) After the BP decoding, the *a-posteriori* LLRs $\mathcal{L}^{\mathrm{p}}_{(t)} = \{\mathbf{l}^{\mathrm{p}}_{(t,1)}, \mathbf{l}^{\mathrm{p}}_{(t,2)}, \ldots, \mathbf{l}^{\mathrm{p}}_{(t,B)}\}$ are repeated to generate $\{\mathcal{L}^{\mathrm{p}}_{(t,r)}\}_{r=2}^{q}$, which are then processed via XOR with the *a-priori* LLRs $\{\mathcal{L}^{\mathrm{h,a}}_{(t,r)}\}_{r=2}^{q}$ obtained from the detection of superposition data at slot t. Subsequently, the XOR yields the interleaved *a-priori* LLRs of Part-3 data at slot $t-1$ (i.e., $\{\hat{\mathcal{L}}^{\mathrm{s,a}}_{(t-1,r)}\}_{r=2}^{q}$) as

$$\hat{\mathbf{l}}^{\mathrm{s,a}}_{(t-1,r,b)}(i) = \log \frac{\exp\left\{\mathbf{l}^{\mathrm{p}}_{(t,r,b)}(i) + \mathbf{l}^{\mathrm{h,a}}_{(t,r,b)}(i)\right\} + 1}{\exp\left\{\mathbf{l}^{\mathrm{p}}_{(t,r,b)}(i)\right\} + \exp\left\{\mathbf{l}^{\mathrm{h,a}}_{(t,r,b)}(i)\right\}}, \tag{26}$$

which is then deinterleaved into $\{\mathcal{L}^{\mathrm{s,a}}_{(t-1,r)}\}_{r=2}^{q}$ and further combined to $\mathcal{L}^{\mathrm{s,a}}_{(t-1)}$.

(ii) The *a-priori* LLRs of both Part-1 and Part-2 data at slot $t-1$ (i.e., $\mathcal{L}^{\mathrm{a}}_{(t-1)}$ and $\mathcal{L}^{\mathrm{a}}_{(t-1,1)}$) are summed with $\mathcal{L}^{\mathrm{s,a}}_{(t-1)}$, and then used for the BP decoding of data at slot $t-1$. The BP decoder outputs the *a-posteriori* LLRs $\mathcal{L}^{\mathrm{p}}_{(t-1)} = \{\mathbf{l}^{\mathrm{p}}_{(t-1,1)}, \mathbf{l}^{\mathrm{p}}_{(t-1,2)}, \ldots, \mathbf{l}^{\mathrm{p}}_{(t-1,B)}\}$.

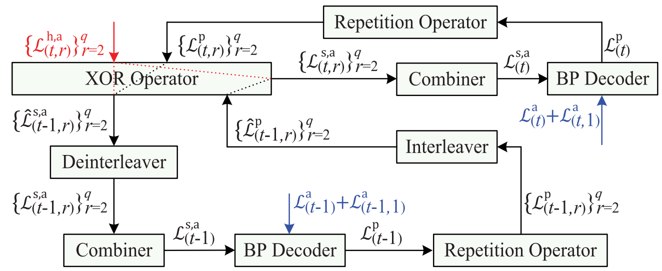


Fig. 5. Mechanism of LLR update in the slot-wise superposition decoding.

(iii) The *a-posteriori* LLRs $\mathcal{L}^{\mathrm{p}}_{(t-1)}$ are repeated to generate $\{\mathcal{L}^{\mathrm{p}}_{(t-1,r)}\}_{r=2}^{q}$, which will be interleaved into $\{\hat{\mathcal{L}}^{\mathrm{p}}_{(t-1,r)}\}_{r=2}^{q}$ and then passed to the XOR module. The XOR can yield the *a-priori* LLRs of Part-3 data at slot t (i.e., $\{\mathcal{L}^{\mathrm{s,a}}_{(t,r)}\}_{r=2}^{q}$) as

$$\mathbf{l}^{\mathrm{s,a}}_{(t,r,b)}(i) = \log \frac{\exp\left\{\hat{\mathbf{l}}^{\mathrm{p}}_{(t-1,r,b)}(i) + \mathbf{l}^{\mathrm{h,a}}_{(t,r,b)}(i)\right\} + 1}{\exp\left\{\hat{\mathbf{l}}^{\mathrm{p}}_{(t-1,r,b)}(i)\right\} + \exp\left\{\mathbf{l}^{\mathrm{h,a}}_{(t,r,b)}(i)\right\}}, \tag{27}$$

which is then combined to $\mathcal{L}^{\mathrm{s,a}}_{(t)}$.

(iv) The *a-priori* LLRs of both Part-1 and Part-2 data at slot t (i.e., $\mathcal{L}^{\mathrm{a}}_{(t)}$ and $\mathcal{L}^{\mathrm{a}}_{(t,1)}$) are summed with $\mathcal{L}^{\mathrm{s,a}}_{(t)}$, and then used for the BP decoding of data at slot t.

Steps (i) $\sim$ (iv) are repeated until the data transmitted at slot $t-1$ is correctly decoded or T_{SD} is reached. To make the above superposition decoding clearer, the corresponding update mechanism of LLRs is presented in Fig. 5.

Remark 2: In the high signal-to-noise ratio (SNR) region, benefiting from the highly reliable information transmission, the proposed SWIT typically does not introduce additional computation overhead compared with the traditional repetition [24]. In the low-to-medium SNR region, the proposed one may involve up to $(3B + qB - 1)n/2$ MPA operations and $2B(T_{\mathrm{SD}} + 1)$ BP operations, while the traditional one requires $(B + qB)n/2$ MPA operations and B BP operations.

V. Proposed Enhanced Multi-User Codes

A. Protograph-Based LDPC Codes

A protograph with a code rate $R = (b_v - b_c)/b_v$ is composed of b_c check nodes (CNs), b_v variable nodes (VNs), and b_e edges connecting both two different types of nodes. Such a protograph can be represented by a base matrix $\mathbf{B} = (b_{i,j})$ of size $b_c \times b_v$, whose rows (resp. columns) correspond to CNs (resp. VNs) and $b_{i,j}$ is the number of edges connecting CN c_i and VN v_j. By exploiting a modified progressive-edge-growth (PEG) algorithm [39], one can extend the base matrix to a parity-check matrix of size $fb_c \times fb_v$, which is used to generate the corresponding rate-R P-LDPC code. In particular, f denotes the lifting factor.

B. Proposed EMUC Design

Due to the inter-user interference, traditional ECCs designed for single-user communications tend to always perform poorly

Algorithm 1 SWIT Receiving Mechanism

Input: The group data $\{\mathcal{G}_1, \mathcal{G}_2, \ldots, \mathcal{G}_{B+1}\}$ and superposition data $\{\mathcal{S}_{(t,2)}, \mathcal{S}_{(t,3)}, \ldots, \mathcal{S}_{(t,q)}\}$ transmitted through a Rayleigh fading channel at slot t. Initialize $\mathbf{l}^{\mathrm{p}}_{(t,1,0)} = \mathbf{0}$ and $\mathcal{F} = 1$.

Implement A-Priori Information Calculation:

Perform MPA on superposition data to get $\{\mathcal{L}^{\mathrm{h,a}}_{(t,r)}\}^{q}_{r=2}$, and then calculate $\{\mathcal{L}^{\mathrm{s,a}}_{(t,r)}\}^{q}_{r=2}$, which can be further combined to generate $\mathcal{L}^{\mathrm{s,a}}_{(t)} = \{\mathbf{l}^{\mathrm{s,a}}_{(t,b)}\}^{B}_{b=1}$.

Implement Group-Wise Sequential Decoding:

for $g = 1, 2, \ldots, B+1$ **do**

  Set $b = g$.

  if $\mathcal{F} = 1$ **then**

    Perform MPA on $\mathcal{G}_g$ with $\mathbf{l}^{\mathrm{p}}_{(t,1,b-1)}$ to get $\mathbf{l}^{\mathrm{a}}_{(t,b)}$.

  Set $\mathcal{F} = 1$.

  Perform BP on $\mathbf{c}_{(t,b)}$ with $\mathbf{l}^{\mathrm{a}}_{(t,b)}$ and $\mathbf{l}^{\mathrm{s,a}}_{(t,b)}$ to get $\mathbf{l}^{\mathrm{p}}_{(t,b)}$.

  if *BP Success* **then**

    Set $\mathbf{l}^{\mathrm{p}}_{(t,1,b)} = \mathbf{l}^{\mathrm{p}}_{(t,b)}$, and terminate current iteration.

  if *BP Failure* **then**

    Perform MPA on $\mathcal{G}_{b+1}$ with $\mathbf{l}^{\mathrm{p}}_{(t,b)}$ to get $\mathbf{l}^{\mathrm{a}}_{(t,1,b)}$ and $\mathbf{l}^{\mathrm{a}}_{(t,b+1)}$, and then perform BP on $\mathbf{c}_{(t,b)}$ with $\mathbf{l}^{\mathrm{a}}_{(t,b)}$, $\mathbf{l}^{\mathrm{s,a}}_{(t,b)}$, and $\mathbf{l}^{\mathrm{a}}_{(t,1,b)}$ to get $\mathbf{l}^{\mathrm{p}}_{(t,1,b)}$.

    if *BP Success* **then**

      Terminate current iteration.

    if *BP Failure* **then**

      Set $\mathcal{F} = 0$, and terminate current iteration.

Implement Slot-Wise Superposition Decoding if the data transmitted at slot $t-1$ is not decoded correctly:

for $k = 1, 2, \ldots, T_{\mathrm{SD}}$ **do**

  for $r = 2, 3, \ldots, q$ **do**

    Let $\mathcal{L}^{\mathrm{p}}_{(t,r)}$ be a copy of $\mathcal{L}^{\mathrm{p}}_{(t)}$.

  Given $\{\mathcal{L}^{\mathrm{p}}_{(t,r)}\}^{q}_{r=2}$, $\{\mathcal{L}^{\mathrm{h,a}}_{(t,r)}\}^{q}_{r=2}$, compute $\{\hat{\mathcal{L}}^{\mathrm{s,a}}_{(t-1,r)}\}^{q}_{r=2}$, which is deinterleaved and further combined to $\mathcal{L}^{\mathrm{s,a}}_{(t-1)}$.

  • Perform BP on the data transmitted at slot $t-1$ with the sum of $\mathcal{L}^{\mathrm{a}}_{(t-1)}$, $\mathcal{L}^{\mathrm{a}}_{(t-1,1)}$, and $\mathcal{L}^{\mathrm{s,a}}_{(t-1)}$ to get $\mathcal{L}^{\mathrm{p}}_{(t-1)}$.

  if *BP Success* **then**

    Break.

  for $r = 2, 3, \ldots, q$ **do**

    Let $\mathcal{L}^{\mathrm{p}}_{(t-1,r)}$ be a copy of $\mathcal{L}^{\mathrm{p}}_{(t-1)}$, and generate its interleaved version $\hat{\mathcal{L}}^{\mathrm{p}}_{(t-1,r)}$.

  Given $\{\hat{\mathcal{L}}^{\mathrm{p}}_{(t-1,r)}\}^{q}_{r=2}$, $\{\mathcal{L}^{\mathrm{h,a}}_{(t,r)}\}^{q}_{r=2}$, compute $\{\mathcal{L}^{\mathrm{s,a}}_{(t,r)}\}^{q}_{r=2}$, which is then combined to $\mathcal{L}^{\mathrm{s,a}}_{(t)}$.

  • Perform BP on the data transmitted at slot t with the sum of $\mathcal{L}^{\mathrm{a}}_{(t)}$, $\mathcal{L}^{\mathrm{a}}_{(t,1)}$, and $\mathcal{L}^{\mathrm{s,a}}_{(t)}$ to get $\mathcal{L}^{\mathrm{p}}_{(t)}$.

Output: Decoding results of data at slows $t-1$ and t.

in SCMA systems [11]. Also, most existing EXIT-based P-LDPC-code design methodologies, including those involving SCMA [11], [27], focus on either non-turbo or turbo decoding. Consequently, these design methodologies typically employ an exhaustive search to find the globally optimal P-LDPC code (i.e., with the lowest decoding threshold) for a single decoding scenario. Obviously, these resultant P-LDPC codes are not general for different decoding paradigms. Therefore, we propose a decoding-threshold-guided multi-objective optimization (DT-MOO) to design enhanced multi-user codes (EMUCs) with robust performance under both non-turbo and turbo decoding scenarios, effectively enabling improved reliability and flexibility in practical SCMA applications.

The proposed DT-MOO then aims to search an EMUC with desirable non-turbo and turbo decoding thresholds (i.e., τ_1 and τ_2), and thus can be expressed as a function $\min\{J_1 = \tau_1, J_2 = \tau_2\}$,[4] where J_1 and J_2 are the objectives that need to be minimized. During the DT-MOO process, there is no a globally optimal solution, and it is essential to find solutions satisfying Pareto optimality (i.e., achieve a balance between J_1 and J_2). The concept of Pareto optimality is described as

- *Definition 1*: A solution $\mathbf{x}_1$ is superior to another solution $\mathbf{x}_2$ if $\mathbf{x}_1$ satisfies the conditions $J_i(\mathbf{x}_1) < J_i(\mathbf{x}_2)$, $\forall i \in \{1, 2\}$;
- *Definition 2*: A solution $\mathbf{x}^*$ is called the Pareto optimal solution if there is no a solution $\mathbf{x}$ being superior to $\mathbf{x}^*$;
- *Definition 3*: All Pareto optimal solutions constitute the Pareto optimal set $\mathbf{X}^*$, and its mapping in objective space is known as the Pareto front, denoted as $P_f = \{J(\mathbf{x}^*) : \mathbf{x}^* \in \mathbf{X}^*\}$.

Algorithm 2 MODE for EMUC Design

Input: $b_{\max}$, b_c, b_v, N_p, $G_{\max}$, P_m, and P_c.

Initialization: Calculate $D = b_c \times b_v$, and produce $\mathbf{P}_1 = \{\mathbf{b}_{1,1}, \mathbf{b}_{1,2}, \ldots, \mathbf{b}_{1,N_p}\}$ by randomly generating one-dimensional vector representation of base matrices $\mathbf{b}_{1,i} = \{b^1_{1,i}, b^2_{1,i}, \ldots, b^D_{1,i}\}$ for $i = 1, 2, \ldots, N_p$.

Compute fitness values (i.e., τ_1 and τ_2) for all base matrices corresponding to the individuals in $\mathbf{P}_1$.

for $g = 1, 2, \ldots, G_{\max}$ **do**

  Mutation: Produce mutant vectors $\mathbf{m}_{g,i}$ for $i = 1, 2, \ldots, N_p$; $\mathbf{m}_{g,i} = \mathbf{b}_{g,\mathrm{r}1} + P_m \times (\mathbf{b}_{g,\mathrm{r}2} - \mathbf{b}_{g,\mathrm{r}3})$, where $\mathbf{b}_{g,\mathrm{r}1}$, $\mathbf{b}_{g,\mathrm{r}2}$, and $\mathbf{b}_{g,\mathrm{r}3}$ are randomly selected from $\mathbf{P}_g$. If the element $m^j_{g,i}$ $(j = 1, 2, \ldots, D)$ within $\mathbf{m}_{g,i}$ is not an integer or exceeds $b_{\max}$, it must be rounded to the nearest integer within the set $\{0, 1, \ldots, b_{\max}\}$.

  Crossover: Produce trial vectors $\mathbf{v}_{g,i} = \{v^1_{g,i}, v^2_{g,i}, \ldots, v^D_{g,i}\}$ for $i = 1, 2, \ldots, N_p$. To be specific, $v^j_{g,i} = m^j_{g,i}$ with probability P_c; Otherwise, $v^j_{g,i} = b^j_{g,i}$, for $j = 1, 2, \ldots, D$.

  Selection: Compute fitness values for all base matrices within $\mathbf{v}_{g,i}$ for $i = 1, 2, \ldots, N_p$, and then perform the following update procedure:

  (i) Update $\mathbf{b}_{g,i} = \mathbf{v}_{g,i}$ if $\mathbf{v}_{g,i}$ dominates $\mathbf{b}_{g,i}$;

  (ii) Remain $\mathbf{b}_{g,i} = \mathbf{b}_{g,i}$ if $\mathbf{b}_{g,i}$ dominates $\mathbf{v}_{g,i}$;

  (iii) Add $\mathbf{v}_{g,i}$ in the population if $\mathbf{v}_{g,i}$ and $\mathbf{b}_{g,i}$ are non-dominated with each other.

  From a population of size ranging from N_p to $2N_p$, one can find the best N_p individuals using a fast non-dominated sorting to constitute $\mathbf{P}_{g+1}$.

Output: $\mathbf{X}^*$, P_f.

To solve the DT-MOO problem, we adopt the multi-objective differential evolution (MODE) to search the Pareto-dominant EMUC base matrix $\mathbf{B}$ (see *Algorithm 2*). The MODE is briefly introduced as follows.

1) Parameter Setting: Input EMUC parameters, including b_c, b_v, and the maximum number of elements $b_{\max}$, and

[4]The decoding thresholds τ_1 and τ_2 are regarded as the minimum SNRs that allow the P-LDPC code to achieve error-free transmission in the SCMA system under non-turbo and turbo decoding scenarios, respectively.

TABLE I

DECODING THRESHOLDS (I.E., τ_1 AND τ_2) OF THE RATE-1/2 P-LDPC CODES (I.E., REGULAR-$(3,6)$, RJA, AR4JA, EARA, TU, AND PROPOSED EMUC-1 CODES) AND THE RATE-2/3 P-LDPC CODES (I.E., AR4JA, EARA, AND PROPOSED EMUC-2 CODES) IN THE 4×6 SCMA SYSTEM OVER A RAYLEIGH FADING CHANNEL. PARTICULARLY, THE PROPOSED NON-UNIFORM CB-1 IS CONSIDERED

$R = 1/2$	Regular-$(3,6)$ [40]	RJA [39]	AR4JA [33]	EARA [41]	TU [42]	Prop. EMUC-1
τ_1 (No Outer Iteration)	4.29	4.23	3.90	4.25	3.93	3.53
τ_2 (With Outer Iterations)	3.26	3.16	3.53	3.22	2.91	2.77

$R = 2/3$	AR4JA [33]	EARA [41]	Prop. EMUC-2
τ_1 (No Outer Iteration)	5.63	5.74	5.53
τ_2 (With Outer Iterations)	5.06	4.61	4.59

MODE parameters, including population size N_p, maximum iterations $G_{\max}$, mutation probability P_m, and crossover probability P_c.

2) Population Initialization: Generate $N_p b_c \times b_v$ base matrices, and then vectorize each of them to a vector representation, thereby formulating an initial population $\mathbf{P}_1$.

3) Mutation, Crossover, and Selection: The individuals within $\mathbf{P}_g$ $(g = 1, 2, \ldots, G_{\max})$ undergo mutation with P_m, followed by crossover with P_c to form a trial population $\mathbf{V}_g$. Then, N_p superior individuals selected from both $\mathbf{P}_g$ and $\mathbf{V}_g$ form next population $\mathbf{P}_{g+1}$. This process repeats until $G_{\max}$ is reached.

For an individual within $\mathbf{P}_1$ and $\mathbf{V}_g$ generated in the MODE procedure, we use Monte-Carlo simulation to obtain the extrinsic LLRs output from SCMA detector, and then compute the corresponding extrinsic mutual information (MI) based on a well-proven LLR-to-MI conversion function [32]. Such an extrinsic MI will be exploited to derive the corresponding decoding threshold by the EXIT analysis [25], [32]. In particular, the Monte-Carlo simulation involves the SCMA property and the Rayleigh fading channel characteristic. In this work, we construct a rate-$1/2$ EMUC of size 4×7 and a rate-$2/3$ EMUC of size 4×10, while both of them have a punctured highest-degree VN. To effectively reduce the search space of EMUCs during MODE procedure, several well-proven code constraints are adopted as[5]

$$\begin{cases} b_{i,j} \in \{0, 1, 2\}, \ \forall\, i = 1, \ldots, b_c \text{ and } j = 1, \ldots, b_v, \\ b_{1,1} = 1, \ \text{while } b_{i,1} = 0 \text{ for } i = 2, 3, 4, \\ b_{2,k} = 1, \ \forall\, k = 2, 3, \ \text{while } b_{2,k'} = 0, \ \forall\, k' = 1, 4, \\ b_{3,k} = 1, \ \forall\, k = 3, 4, \ \text{while } b_{3,k'} = 0, \ \forall\, k' = 1, 2, \\ b_{4,k} = 1, \ \forall\, k = 2, 4, \ \text{while } b_{4,k'} = 0, \ \forall\, k' = 1, 3, \\ 3 \le \displaystyle\sum_{i=1}^{4} b_{i,j} \le \sum_{i=1}^{4} b_{i,b_v}, \ \forall\, 4 < j < b_v. \end{cases} \tag{28}$$

Referring to (28), (i) Line 1 restricts the base-matrix elements to a maximum value of 2 to reduce the search space; (ii) Line 2 sets a degree-1 VN, while Lines 3 $\sim$ 5 configures three degree-2 VNs, thereby contributing to an improved decoding threshold; (iii) Line 6 limits the number of VNs with degrees not exceeding 2 to $b_c = 4$, helping avoid the error-floor phenomenon. Exploiting *Algorithm 2* with $N_p = 10$, $G_{\max} = 100$, $P_m = 0.5$, and $P_c = 0.8$, the resultant rate-$1/2$ and rate-$2/3$ EMUCs, called *EMUC-1* and *EMUC-2*, respectively, are obtained as

$$\mathbf{B}_{\text{EMUC-1}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 & 1 & 1 & 2 \\ 0 & 1 & 1 & 0 & 2 & 0 & 2 \\ 0 & 0 & 1 & 1 & 0 & 2 & 1 \end{bmatrix},$$
$$\mathbf{B}_{\text{EMUC-2}} = \begin{bmatrix} 1 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 2 \\ 0 & 1 & 0 & 1 & 1 & 2 & 1 & 0 & 0 & 2 \\ 0 & 1 & 1 & 0 & 2 & 1 & 0 & 2 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 & 0 & 2 & 1 & 2 & 1 \end{bmatrix}. \tag{29}$$

In particular, the VN 1 and VNs 2 to 4 correspond to the predefined degree-1 VN and degree-2 VNs, respectively, while distributing '1' elements in different rows is to ensure that both EMUC-1 and EMUC-2 achieve low decoding thresholds and avoid the 4-cycle structure. The VN 7 (resp. VN 10) correspond to the punctured highest-degree VN of EMUC-1 (resp. EMUC-2).

C. Decoding-Threshold Analysis

To validate the superiorities of the proposed rate-$1/2$ and rate-$2/3$ EMUCs in SCMA systems over a Rayleigh fading channel, their non-turbo and turbo decoding thresholds (i.e., τ_1 and τ_2) are presented in Table I. In particular, the proposed 4×6 non-uniform CB-1 is considered. Also, the rate-$1/2$ regular-$(3,6)$ [40], repeat-jagged-accumulate (RJA) [39], accumulate-repeat-by-4-jagged-accumulate (AR4JA) [33], enhanced accumulate-repeat-accumulate (EARA) [41], and TU [42] P-LDPC codes, as well as the rate-$2/3$ AR4JA [33] and EARA [41] P-LDPC codes, are used as benchmarks. Additionally, the error-correction-code parameters and receiver parameters are assumed as (i) The information length is 1200; (ii) In the case of non-turbo decoding (i.e., no outer iteration), T_{MPA} and T_{BP} are set to 10 and 100, respectively; (iii) In the case of turbo decoding (i.e., with outer iteration), T_{out}, T_{MPA}, and T_{BP} are set to 4, 5, and 50, respectively.

As observed from this table, in the case of $R = 1/2$, the proposed EMUC-1 significantly outperform the other five different P-LDPC codes. Specifically, the proposed EMUC-1 exhibits the smallest decoding thresholds 3.53 dB and 2.77 dB under non-turbo and turbo decoding scenarios, respectively, while the regular-$(3,6)$, RJA, AR4JA, EARA, and TU P-LDPC codes require much larger decoding thresholds. Furthermore, a similar trend in decoding-threshold comparison can be observed among the rate-$2/3$ AR4JA, EARA, and EMUC-2 codes. Notably, the proposed EMUCs achieve optimal convergence performance under both non-turbo and turbo

[5]In this work, we initialize a small-size base matrix and some constraints to lower the EMUC construction complexity. However, the EMUC with better performance can be constructed if DT-MOO is used to search a large-size base matrix without considering any constraint.

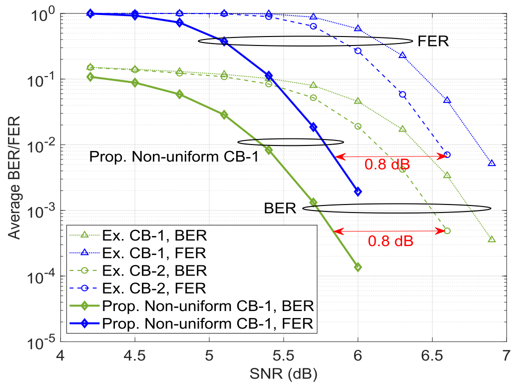


Fig. 6. Average BER/FER performance curves of the existing CB-1 [12], CB-2 [35], and proposed non-uniform CB-1 in 4×6 SCMA system. In particular, the regular-$(3, 6)$ P-LDPC code is considered, while the parameters are set to $T_{\mathrm{MPA}} = 10$ and $T_{\mathrm{BP}} = 100$ (i.e., non-turbo decoding).

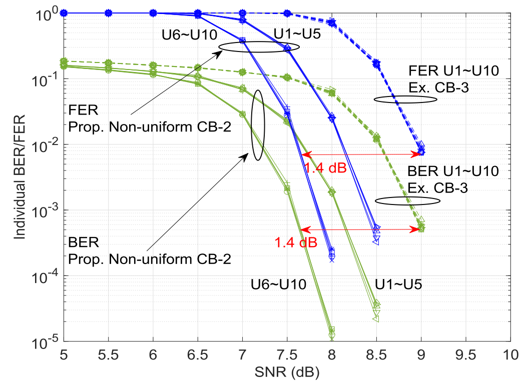


Fig. 7. Individual BER/FER performance curves of the existing CB-3 [12] and proposed non-uniform CB-2 in 5×10 SCMA system. In particular, the regular-$(3, 6)$ P-LDPC code is considered, while the parameters are set to $T_{\mathrm{MPA}} = 10$ and $T_{\mathrm{BP}} = 100$ (i.e., non-turbo decoding).

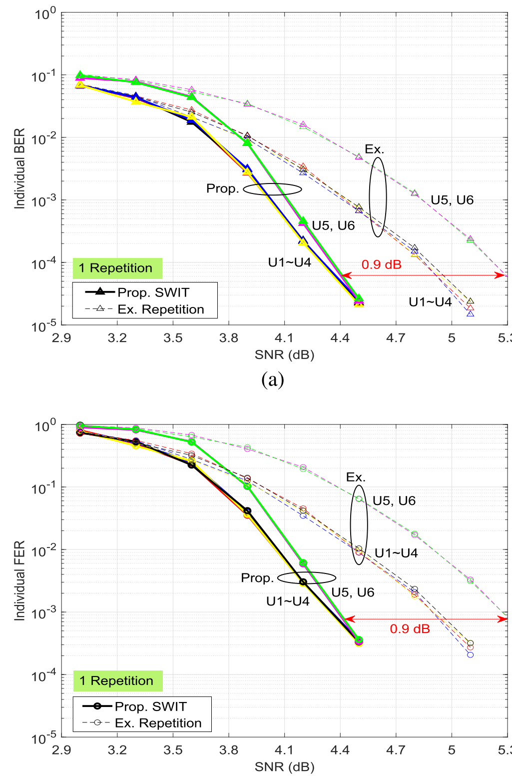
decoding, whereas traditional P-LDPC codes may not achieve this consistency. For example, the AR4JA code outperforms the regular-$(3, 6)$, RJA, EARA, and TU P-LDPC codes under non-turbo decoding, but it exhibits the worst convergence performance under turbo decoding.

*N*ote also that:

- Considering the error-rate simulation, where a finite-length P-LDPC code is assumed, the decoding threshold is used to predict its waterfall-region performance. For different P-LDPC codes, we can compare their decoding thresholds to evaluate their convergence performance. Importantly, the gap between the decoding threshold and the finite-length error-rate performance can be narrowed by increasing the transmitted codeword length.
- The EXIT [33] and density-evolution [43] algorithms are the two most popular tools for LDPC-coded optimization under iterative decoding. However, the former can achieve a comparable accuracy as the latter in decoding-threshold analysis with a relatively lower complexity. Owing to this inherent advantage of the EXIT algorithm, various EXIT-aided analytical methods have been developed to facilitate the design of LDPC-coded systems [27], [39], [41]. Therefore, we also employ the EXIT algorithm to analyze the decoding thresholds of the proposed SCMA system.
- During the decoding-threshold-based EMUC design process, the channel is modeled with the Rayleigh fading, perfect channel state information (CSI), and coherence time lasting for one transmitted modulation symbol.

VI. Simulation Results

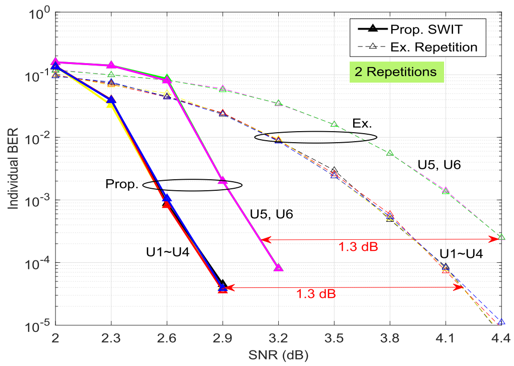
To validate the superiorities of the proposed designs, this section presents various error-rate simulations on P-LDPC-coded SCMA systems over a Rayleigh fading channel. Specifically, two existing CB-1 [12], CB-2 [35], and proposed non-uniform CB-1 are used for 4×6 SCMA system, while the existing CB-3 [12] and proposed non-uniform CB-2 are used for 5×10 SCMA system. Moreover, the traditional repetition scheme [24] and proposed SWIT scheme are considered. Furthermore, different channel coding schemes are adopted, including (i) the rate-$1/2$ regular-$(3, 6)$ [40], RJA [39], AR4JA [33], EARA [41], 5G NR [44], GMAC [26], GMU [45], TU [42], and proposed EMUC-1 codes; and (ii) the rate-$2/3$ AR4JA [33], EARA [41], 5G NR [44], and proposed EMUC-2 codes. Besides, it should be noted that the non-turbo decoding (i.e., $T_{\mathrm{MPA}} = 10$ and $T_{\mathrm{BP}} = 100$) is considered unless the turbo decoding (i.e., $T_{\mathrm{out}} = 4$, $T_{\mathrm{MPA}} = 5$, and $T_{\mathrm{BP}} = 50$) is emphasized. For the SWIT scheme, the number of coded blocks transmitted per slot before repetition is set to $B = 20$.



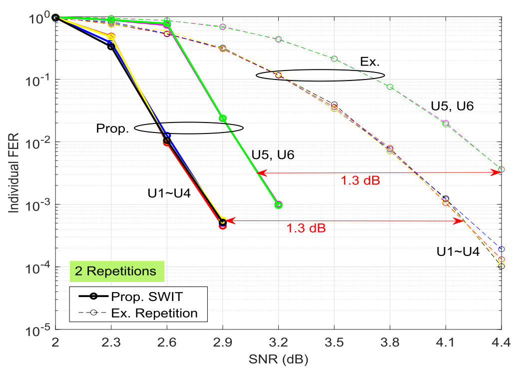
(a)

(b)

Fig. 8. Performance comparison between the existing repetition scheme [24] and proposed SWIT scheme in 4×6 SCMA system: (a) individual BER performance curves; and (b) individual FER performance curves. In particular, the proposed non-uniform CB-1 and regular-$(3, 6)$ P-LDPC code are considered, with parameters set to $q = 1$, $T_{\mathrm{MPA}} = 10$, and $T_{\mathrm{BP}} = 100$ (i.e., non-turbo decoding).

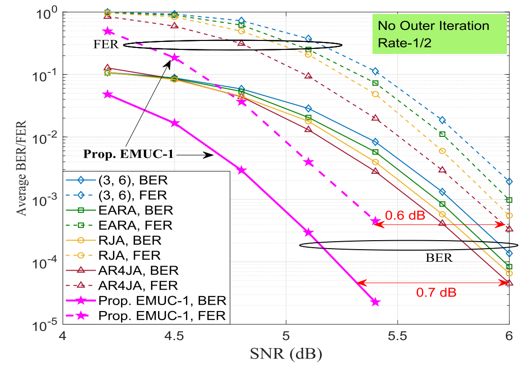


(a)

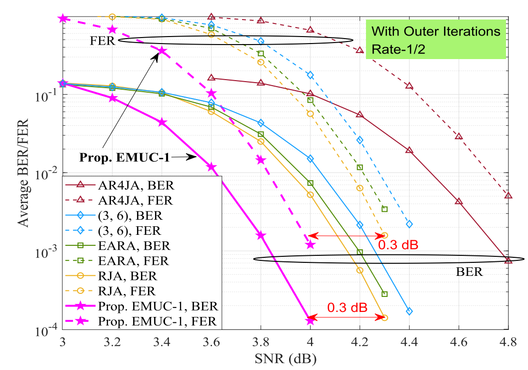


(b)

Fig. 9. Performance comparison between the existing repetition scheme [24] and proposed SWIT scheme in 4×6 SCMA system: (a) individual BER performance curves; and (b) individual FER performance curves. In particular, the proposed non-uniform CB-1 and regular-$(3, 6)$ P-LDPC code are considered, with parameters set to $q = 2$, $T_{\mathrm{MPA}} = 10$, and $T_{\mathrm{BP}} = 100$ (i.e., non-turbo decoding).



(a)



(b)

Fig. 10. Average BER/FER performance curves of the traditional rate-$1/2$ single-user P-LDPC codes (i.e., regular-$(3, 6)$ [40], RJA [39], AR4JA [33], and EARA [41]) and proposed EMUC-1 code in 4×6 SCMA system with proposed non-uniform CB-1: (a) the parameters $T_{\mathrm{MPA}} = 10$ and $T_{\mathrm{BP}} = 100$ are considered (i.e., non-turbo decoding); (b) the parameters $T_{\mathrm{out}} = 4$, $T_{\mathrm{MPA}} = 5$, and $T_{\mathrm{BP}} = 50$ are considered (i.e., turbo decoding).

A. Performance of the Proposed Non-Uniform Codebooks

Fig. 6 plots the average bit-error-rate (BER) / frame-error-rate (FER) curves of the existing CB-1 [12], CB-2 [35], and proposed non-uniform CB-1 in the 4×6 SCMA system with regular-$(3, 6)$ P-LDPC code under non-turbo decoding. Referring to this figure, at a BER of 5×10^{-4} (resp. FER of 6×10^{-3}), one can easily observe that the proposed non-uniform CB-1 achieves BER (resp. FER) performance gains of around 0.8 dB and 1.1 dB over the existing CB-2 and CB-1, respectively.

Fig. 7 plots the individual BER/FER curves of the existing CB-3 [12] and proposed non-uniform CB-2 in the 5×10 SCMA system with regular-$(3, 6)$ P-LDPC code under non-turbo decoding. As illustrated in this figure, the proposed non-uniform CB-2 significantly outperforms the existing CB-3 in both of BER and FER. Specifically, at an SNR of 8.0 dB, the proposed non-uniform CB-2 can make UEs $6 \sim 10$ and UEs $1 \sim 5$ achieve BERs of 1×10^{-5} and 2×10^{-3}, respectively, while all users with the existing CB-3 can only achieve a BER of 6×10^{-2}. Furthermore, one can observe a similar performance trend in term of FER.

B. Performance of the Proposed SWIT Scheme

Fig. 8 studies the BER/FER performance of 4×6 SCMA system with proposed SWIT scheme under non-turbo decoding. The existing repetition scheme [24] is used as benchmark, while the proposed non-uniform CB-1, regular-$(3, 6)$ P-LDPC code, and $q = 1$ are considered. Referring to Fig. 8(a), one can easily find that the proposed SWIT scheme achieves a maximum gain of around 0.9 dB over the existing repetition scheme at a BER of 6×10^{-5}. Moreover, at an SNR of 4.5 dB, all users with the proposed SWIT scheme achieve a BER of 2×10^{-5}, while the UEs $1 \sim 4$ and UEs $5, 6$ with the existing repetition scheme only achieve BERs of 6×10^{-4} and 5×10^{-3}, respectively. Additionally, from the perspective of FER performance shown in Fig. 8(b), a similar convergence performance trend can be observed.

Similarly, Fig. 9 analyzes the BER/FER performance of both proposed SWIT scheme and existing repetition scheme under the same simulation conditions except that $q = 2$. Referring to Fig. 9(a), it is obvious that compared with the existing repetition scheme, the UEs $1 \sim 4$ (resp. UEs $5, 6$) with the proposed SWIT scheme achieve a BER performance gain of around 1.3 dB at a BER of 4×10^{-5} (resp. 2×10^{-3}). Furthermore, as depicted in Fig. 9(b), one can observe a similar FER performance trend. Note that since the parameters are set to $k = 500$, $n = 1000$, $B = 20$ and $q = 2$, the transmission rate loss caused by the proposed SWIT scheme is then 0.0027 compared with the repetition scheme.

C. Performance of the Proposed EMUCs

Fig. 10 investigates the average BER/FER performance of the rate-$1/2$ single-user P-LDPC codes (i.e., regular-$(3, 6)$

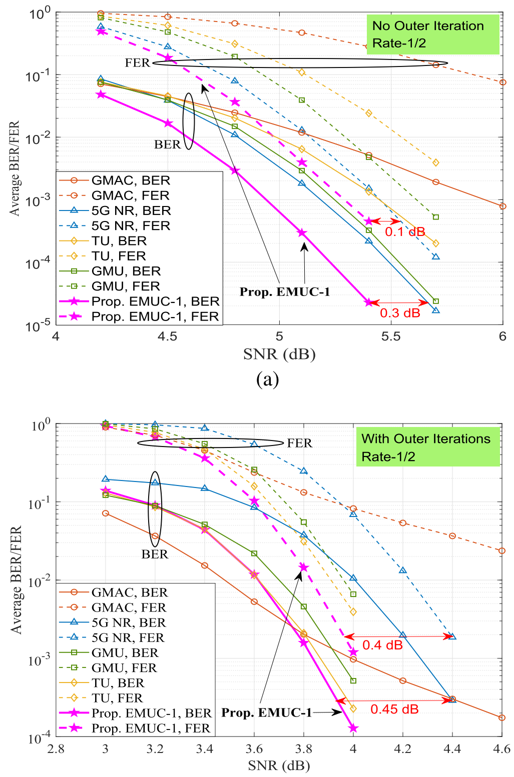


Fig. 11. Average BER/FER performance curves of the existing rate-$1/2$ 5G NR [44] code, multi-user P-LDPC codes (i.e., GMAC [26], GMU [45], and TU [42]) and proposed EMUC-1 code in 4×6 SCMA system with proposed non-uniform CB-1: (a) the parameters $T_{\mathrm{MPA}} = 10$ and $T_{\mathrm{BP}} = 100$ are considered (i.e., non-turbo decoding); (b) the parameters $T_{\mathrm{out}} = 4$, $T_{\mathrm{MPA}} = 5$, and $T_{\mathrm{BP}} = 50$ are considered (i.e., turbo decoding).

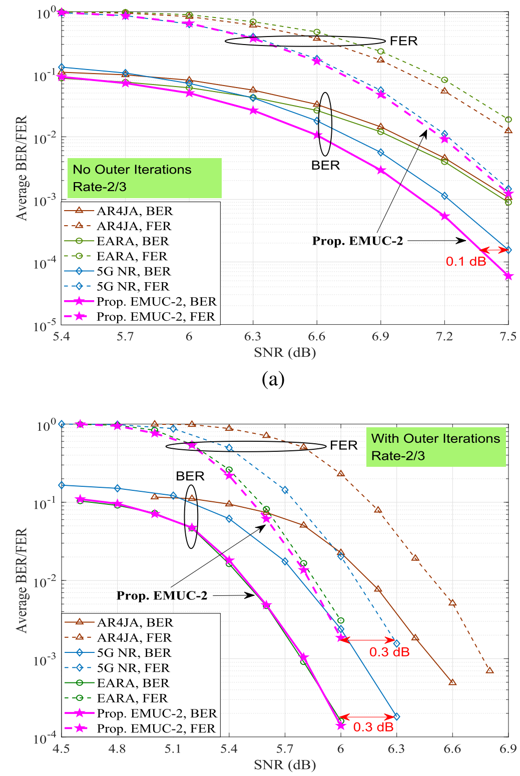


Fig. 12. Average BER/FER performance curves of the existing rate-$2/3$ single-user P-LDPC codes (i.e.,, AR4JA [33], EARA [41], and 5G NR [44]) and proposed EMUC-2 code in 4×6 SCMA system with proposed non-uniform CB-1: (a) the parameters $T_{\mathrm{MPA}} = 10$ and $T_{\mathrm{BP}} = 100$ are considered (i.e., non-turbo decoding); (b) the parameters $T_{\mathrm{out}} = 4$, $T_{\mathrm{MPA}} = 5$, and $T_{\mathrm{BP}} = 50$ are considered (i.e., turbo decoding).

[40], RJA [39], AR4JA [33], and EARA [41] codes) and proposed EMUC-1 code in 4×6 SCMA system with proposed non-uniform CB-1. As observed from Fig. 10(a), without considering outer iteration, the proposed EMUC-1 code significantly outperforms the other four different single-user P-LDPC codes in both BER and FER. Specifically, at a BER of 1×10^{-4} (resp. FER of 4×10^{-3}), the proposed EMUC-1 code achieves a BER (resp. FER) performance gain of about 0.7 dB (resp. 0.6 dB) over the AR4JA code, while the latter is already the best-performing code among the four single-user P-LDPC codes. Similarly, referring to Fig. 10(b), in which the outer iterations are considered, the proposed EMUC-1 code can still exhibit better performance in both BER and FER compared to the other four single-user P-LDPC codes.

Fig. 11 further compares the average BER/FER performance of the existing rate-$1/2$ 5G NR [44] code, multi-user P-LDPC codes (i.e., GMAC [26], GMU [45], and TU [42] codes), and proposed EMUC-1 code in 4×6 SCMA system with proposed non-uniform CB-1. As seen from Fig. 11(a), without considering outer iteration, the proposed EMUC-1 code achieves better BER and FER performance than the four different existing P-LDPC codes. Specifically, at an SNR of 5.4 dB, the proposed EMUC-1 code can achieve a BER of 2×10^{-5} (resp. FER of 4×10^{-4}), while the 5G NR, GMU, TU, and GMAC codes only accomplish BERs of 2×10^{-4}, 3×10^{-4}, 1.5×10^{-3}, and 5×10^{-3} (resp. FERs of 1.5×10^{-3}, 5×10^{-3}, 2×10^{-2}, and 3×10^{-1}), respectively. On the other hand, from Fig. 11(b), even with outer iterations, one can notice that the proposed EMUC-1 code can still perform better than the existing 5G NR code and multi-user P-LDPC codes.

Fig. 12 investigates the average BER/FER performance of the existing rate-$2/3$ single-user P-LDPC codes (i.e., AR4JA [33], EARA [41], and 5G NR [44] codes) and proposed EMUC-2 code in 4×6 SCMA system with proposed non-uniform CB-1. As observed from Fig. 12(a), without considering outer iteration, the proposed EMUC-2 code can not only exhibit the best BER performance, but also achieve comparable FER performance with respect to the 5G NR code, while the latter performs better than both EARA and AR4JA codes across the entire SNR range. Additionally, referring to Fig. 12(b) in which outer iterations are considered, the proposed EMUC-2 code possesses the same BER and FER performance as EARA code. Notably, the EARA code achieves BER (resp. FER) performance gains of around 0.3 dB and 0.7 dB over the 5G NR and AR4JA codes, respectively, at a BER of 5×10^{-4} (resp. FER of 2×10^{-3}).

Remark 3: As observed from Figs. 10 $\sim$ 12, one can notice that (i) The prior-art P-LDPC codes fail to simultaneously achieve reliable error-rate performance under both non-turbo decoding and turbo decoding. Moreover, the punctured P-LDPC codes (including AR4JA, EARA, and 5G NR codes) exhibit worse performance under outer iterations compared to the unpunctured counterpart (i.e., RJA code). This observation is quite different from that in the traditional single-user scenario; (ii) The prior-art P-LDPC codes cannot simultaneously achieve excellent BER and FER performance, as exemplified by the AR4JA and EARA codes. To address these two limitations, the proposed EMUCs can simultaneously achieve superior non-turbo and turbo decoding performance, in terms of BER and FER.

VII. Conclusion

This paper has investigated the design of P-LDPC-coded SCMA systems with repetition-based transmission over Rayleigh fading channels. We have first proposed a two-step design method, which consists of LDFG and ECS principles, to construct a new type of non-uniform CBs with stronger robustness against inter-user interference compared with existing CBs. The LDFG principle is used to determine the non-uniform mapping relationship between UEs and REs, while the ECS principle is used to determine the resultant multi-user symbol superposition in RE domain. To fully leverage the repetition-based transmission, we have introduced a novel SWIT scheme, featuring a systematic optimization among repetition coding, resource mapping, and cross-slot data coupling for enhancing SCMA robustness. As a further advancement, we have conceived a DT-MOO method to design EMUCs. The proposed EMUCs can achieve better error-rate performance in SCMA systems under both turbo and non-turbo decoding, with respect to the traditional single-user and existing multi-user codes. Numerical results confirm the superior performance of proposed designs, highlighting their potential as promising alternatives to benchmark schemes. Hence, the proposed designs appear to be a prospective technique for the energy-constrained scenarios with ultra-dense connectivity requirements, especially emerging SWIPT-enabled 6G networks.

References

[1] G. Pan, H. Zhang, R. Zhang, S. Wang, J. An, and M.-S. Alouini, "Space simultaneous information and power transfer: An enhanced technology for miniaturized satellite systems," *IEEE Wireless Commun.*, vol. 30, no. 2, pp. 122–129, Apr. 2023.

[2] X. Zhang et al., "GNN-based secrecy rate optimization in multi-satellite collaborative systems," *IEEE J. Sel. Areas Commun.*, vol. 44, pp. 3648–3663, 2026.

[3] S. Wang et al., "GaussMask-DSSS: Enhancing covert spread spectrum communication with Gaussian cloaking and deep learning-aided synchronization," *IEEE J. Sel. Areas Commun.*, vol. 44, pp. 1495–1511, 2026.

[4] Y. Fang, J. Yang, D. Ma, M. Yang, Z. Xu, and X. Chen, "Integrated sensing and backscatter communication with movable antennas: State-of-the-art survey and a novel inverse scattering framework," *IEEE Trans. Netw. Sci. Eng.*, vol. 13, pp. 4350–4368, 2026.

[5] H. Xu, C. Zhang, Q. Wu, B. K. Ng, C.-T. Lam, and H. Yanikomeroglu, "FTN-assisted SWIPT-NOMA design for IoT wireless networks: A paradigm in wireless efficiency and energy utilization," *IEEE Sensors J.*, vol. 25, no. 4, pp. 7431–7444, Feb. 2025.

[6] A. Al-Fuqaha, M. Guizani, M. Mohammadi, M. Aledhari, and M. Ayyash, "Internet of Things: A survey on enabling technologies, protocols, and applications," *IEEE Commun. Surveys Tuts.*, vol. 17, no. 4, pp. 2347–2376, 4th Quart., 2015.

[7] Q. Luo, Z. Liu, G. Chen, P. Xiao, Y. Ma, and A. Maaref, "A design of low-projection SCMA codebooks for ultra-low decoding complexity in downlink IoT networks," *IEEE Trans. Wireless Commun.*, vol. 22, no. 10, pp. 6608–6623, Oct. 2023.

[8] S. M. R. Islam, N. Avazov, O. A. Dobre, and K.-S. Kwak, "Power-domain non-orthogonal multiple access (NOMA) in 5G systems: Potentials and challenges," *IEEE Commun. Surveys Tuts.*, vol. 19, no. 2, pp. 721–742, 2nd Quart., 2017.

[9] Z. Liu and L.-L. Yang, "Sparse or dense: A comparative study of code-domain NOMA systems," *IEEE Trans. Wireless Commun.*, vol. 20, no. 8, pp. 4768–4780, Aug. 2021.

[10] Q. Luo, Z. Liu, G. Chen, and P. Xiao, "Enhancing signal space diversity for SCMA over Rayleigh fading channels," *IEEE Trans. Wireless Commun.*, vol. 23, no. 4, pp. 3676–3690, Apr. 2024.

[11] J. Chen, Z. Yang, Y. Fang, Y. Li, and J. Zhan, "Design of protograph LDPC-coded SCMA systems with power-imbalanced multi-level decoding," *IEEE Trans. Veh. Technol.*, vol. 75, no. 1, pp. 1643–1648, Jan. 2026.

[12] (2015). *Altera Innovate Asia Website, Presentation*. [Online]. Available: http://www.innovateasia.com/5g/en/gp2.html

[13] C. Yan, G. Kang, and N. Zhang, "A dimension distance-based SCMA codebook design," *IEEE Access*, vol. 5, pp. 5471–5479, 2017.

[14] M. Taherzadeh, H. Nikopour, A. Bayesteh, and H. Baligh, "SCMA codebook design," in *Proc. IEEE 80th Veh. Technol. Conf. (VTC-Fall)*, Sep. 2014, pp. 1–5.

[15] X. Li, Z. Gao, Y. Gui, Z. Liu, P. Xiao, and L. Yu, "Design of power-imbalanced SCMA codebook," *IEEE Trans. Veh. Technol.*, vol. 71, no. 2, pp. 2140–2145, Feb. 2022.

[16] S. Zhang et al., "A capacity-based codebook design method for sparse code multiple access systems," in *Proc. 8th Int. Conf. Wireless Commun. Signal Process. (WCSP)*, Oct. 2016, pp. 1–5.

[17] L. Yu, X. Lei, P. Fan, and D. Chen, "An optimized design of SCMA codebook based on star-QAM signaling constellations," in *Proc. Int. Conf. Wireless Commun. Signal Process. (WCSP)*, Oct. 2015, pp. 1–5.

[18] Y. Zhou, Q. Yu, W. Meng, and C. Li, "SCMA codebook design based on constellation rotation," in *Proc. IEEE Int. Conf. Commun. (ICC)*, May 2017, pp. 1–6.

[19] Z. Mheich, L. Wen, P. Xiao, and A. Maaref, "Design of SCMA codebooks based on golden angle modulation," *IEEE Trans. Veh. Technol.*, vol. 68, no. 2, pp. 1501–1509, Feb. 2019.

[20] G. Song, X. Wang, and J. Cheng, "A low-complexity multiuser coding scheme with near-capacity performance," *IEEE Trans. Veh. Technol.*, vol. 66, no. 8, pp. 6775–6786, Aug. 2017.

[21] G. Song, J. Cheng, and Y. Watanabe, "Maximum sum rate of repeat-accumulate interleave-division system by fixed-point analysis," *IEEE Trans. Commun.*, vol. 60, no. 10, pp. 3011–3022, Oct. 2012.

[22] Y. Zhang, W. Hou, Y. Li, and J. Cheng, "Robust rateless spatially coupled repeat-accumulate-repeat multi-user codes on IDMA systems," *IEEE Trans. Veh. Technol.*, vol. 70, no. 11, pp. 11523–11537, Nov. 2021.

[23] Y. Chi, Y. Li, G. Song, and Y. Sun, "Partially repeated SC-LDPC codes for multiple-access channel," *IEEE Commun. Lett.*, vol. 20, no. 10, pp. 1947–1950, Oct. 2016.

[24] K. Lai, Z. Liu, J. Lei, G. Chen, P. Xiao, and L. Wen, "Sparse code multiple access with enhanced K-repetition scheme: Analysis and design," *IEEE Trans. Wireless Commun.*, vol. 23, no. 6, pp. 5301–5315, Jun. 2024.

[25] S. Khittiwitchayakul, W. Phakphisut, P. Supnithi, and L. M. Min Myint, "EXIT chart analysis of protograph-based LDPC codes for unequal power two-user Gaussian multiple access channel," in *Proc. 35th Int. Tech. Conf. Circuits/Syst., Comput. Commun. (ITC-CSCC)*, Jul. 2020, pp. 354–358.

[26] X. Li, P. Chen, Y. Fang, and Z. Xie, "Design of protograph-based LDPC codes for two-user Gaussian multiple access channel," *IEEE Wireless Commun. Lett.*, vol. 11, no. 9, pp. 1990–1994, Sep. 2022.

[27] J. Chen, H. Ma, L. Lv, and W. Xue, "Design of enhanced protograph codes and codebooks for IDD-SCMA systems," *IEEE Access*, vol. 13, pp. 51429–51439, 2025.

[28] L. Liu, C. Liang, J. Ma, and L. Ping, "Capacity optimality of AMP in coded systems," *IEEE Trans. Inf. Theory*, vol. 67, no. 7, pp. 4429–4445, Jul. 2021.

[29] J. R. Ebert, J. Chamberland, and K. R. Narayanan, "Multi-user SR-LDPC codes via coded demixing with applications to cell-free systems," in *Proc. IEEE Int. Symp. Inf. Theory*, Jul. 2024, pp. 2015–2020.

[30] X. Fu, M. Pischella, and D. Le Ruyet, "Joint uplink PD-NOMA and SCMA for future multiple access systems," in *Proc. IEEE 93rd Veh. Technol. Conf. (VTC-Spring)*, Apr. 2021, pp. 1–5.

[31] S. Sharma, K. Deka, V. Bhatia, and A. Gupta, "Joint power-domain and SCMA-based NOMA system for downlink in 5G and beyond," *IEEE Commun. Lett.*, vol. 23, no. 6, pp. 971–974, Jun. 2019.

[32] Z. Yang, Y. Ge, Y. Zhao, Y. Fang, and Y. L. Guan, "Protograph LDPC code and shaped index modulation design for multi-mode OAM systems," *IEEE Trans. Commun.*, vol. 72, no. 8, pp. 5162–5178, Aug. 2024.

[33] Z. Yang, Y. Li, Y. L. Guan, and Y. Fang, "Source-constrained hierarchical modulation systems with protograph LDPC codes: A promising transceiver design for future 6G-enabled IoT," *IEEE J. Sel. Areas Commun.*, vol. 43, no. 4, pp. 1103–1117, Apr. 2025.

[34] X. Ren et al., "A high performance and low complexity LDPC decoding strategy for space-air–ground-sea integrated communication networks," *IEEE Trans. Veh. Technol.*, vol. 74, no. 7, pp. 11461–11465, Jul. 2025.

[35] Y.-M. Chen and J.-W. Chen, "On the design of near-optimal sparse code multiple access codebooks," *IEEE Trans. Commun.*, vol. 68, no. 5, pp. 2950–2962, May 2020.

[36] Y. Zheng, J. Xin, H. Wang, S. Zhang, and Y. Qiao, "A low-complexity codebook design scheme for SCMA systems over an AWGN channel," *IEEE Trans. Veh. Technol.*, vol. 71, no. 8, pp. 8675–8688, Aug. 2022.

[37] D. G. M. Mitchell, M. Lentmaier, and D. J. Costello, "Spatially coupled LDPC codes constructed from protographs," *IEEE Trans. Inf. Theory*, vol. 61, no. 9, pp. 4866–4889, Sep. 2015.

[38] J. Yang, Q. Wang, Z. Tan, C. Li, and X. Ma, "Spatially coupled 5G LDPC codes via superposition," in *Proc. IEEE Wireless Commun. Netw. Conf. (WCNC)*, Mar. 2025, pp. 1–6.

[39] Z. Yang, Y. Fang, G. Zhang, and F. C. M. Lau, "Design and optimization of protograph LDPC-coded multipulse PPM systems over Poisson channels," *IEEE Trans. Veh. Technol.*, vol. 71, no. 9, pp. 9586–9601, Sep. 2022.

[40] Z. Yang, X. Yu, Y. Hu, Y. Fang, and Y. L. Guan, "Nested root-protograph LDPC-coded multilevel-priority IR-HARQ systems: A new design for multi-stream transmissions with non-ergodic block fading," *IEEE Trans. Commun.*, vol. 73, no. 11, pp. 1–15, Nov. 2025.

[41] L. Dai, Y. Fang, Y. L. Guan, and M. Guizani, "Design of protograph LDPC-coded MIMO-VLC systems with generalized spatial modulation," *China Commun.*, vol. 99, no. 99, pp. 1–14, Jan. 2023.

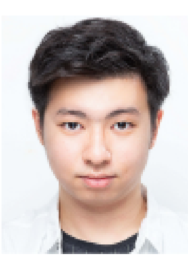
[42] W. Zhang, P. Chen, Z. Xie, and F. Chen, "Design of protograph LDPC codes for the unequal power two-user downlink system," in *Proc. 13th Int. Conf. Wireless Commun. Signal Process. (WCSP)*, Oct. 2021, pp. 1–5.

[43] T. J. Richardson, M. A. Shokrollahi, and R. L. Urbanke, "Design of capacity-approaching irregular low-density parity-check codes," *IEEE Trans. Inf. Theory*, vol. 47, no. 2, pp. 619–637, Feb. 2001.

[44] *5G NR: Multiplexing Channel Coding*, document ETSI TS 138 212, Jul. 2018.

[45] Y. Chi, L. Liu, G. Song, Y. Li, Y. L. Guan, and C. Yuen, "Constrained capacity optimal generalized multi-user MIMO: A theoretical and practical framework," *IEEE Trans. Commun.*, vol. 70, no. 12, pp. 8086–8104, Dec. 2022.

Zhaojie Yang received the Ph.D. degree in communication engineering from Guangdong University of Technology, China, in 2021. From 2022 to 2025, he was a Research Fellow with the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore. He is currently an Associate Professor with the School of Information Engineering, Guangdong University of Technology.



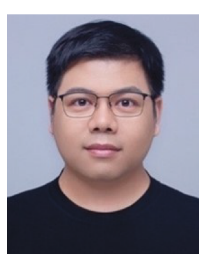
Yunye Li (Graduate Student Member, IEEE) received the M.Sc. degree in information and communication engineering from Guangdong University of Technology, China, in 2023, where he is currently pursuing the Ph.D. degree in information and communication engineering. From 2023 to 2024, he was a Research Associate with Nanyang Technological University, Singapore. His current research interests include protograph LDPC codes and SCMA systems.

Xiaoxi Yu received the B.Eng. degree in electrical and electronics engineering and the M.Sc. degree in wireless communications and signal processing from the University of Bristol, Bristol, U.K., in 2016 and 2018, respectively, and the Ph.D. degree in electrical and electronic engineering from Nanyang Technological University, Singapore, in 2024. She is currently a Research Fellow with Nanyang Technological University.

Jiayi Yang is currently pursuing the Ph.D. degree with Sun Yat-sen University, Guangzhou, China. His research interests include channel coding, coded modulation, waveform designs, and their applications to wireless communication systems.

Yi Fang (Senior Member, IEEE) received the Ph.D. degree in communication engineering, Xiamen University, China, in 2013. From September 2012 to September 2013, he was a Visiting Scholar in electronic and electrical engineering with University College London, U.K. From February 2014 to February 2015, he was a Research Fellow with the School of Electrical and Electronic Engineering, Nanyang Technological University, Singapore. He is currently a Full Professor and the Vice Dean with the School of Information Engineering, Guangdong University of Technology, China.

Yong Liang Guan (Senior Member, IEEE) received the B.Eng. degree (Hons.) from the National University of Singapore, Singapore, and the Ph.D. degree from Imperial College London, London, U.K. He is currently an Associate Vice President with Nanyang Technological University (NTU), Singapore, and a Professor of communication engineering with the School of Electrical and Electronic Engineering, NTU, where he has founded and is leading the Continental-NTU Corporate Research Laboratory. He also led the successful deployment of the campus-wide NTU-NXP V2X Test Bed, NTU. His research interests include coding and signal processing for communication systems and data storage systems. He is an Editor of IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY and a Distinguished Speaker of the IEEE Vehicular Technology Society.



Yuhao Chi (Senior Member, IEEE) received the Ph.D. degree in communication and information systems from Xidian University (XDU), Xi'an, China, in 2018. From 2016 to 2017, he was an Exchange Ph.D. Student with Nanyang Technological University (NTU), Singapore, supported by the State Scholarship Fund from China Scholarship Council and a Visiting Student with Singapore University of Technology and Design (SUTD), Singapore. From 2018 to 2021, he was a Senior Engineer with Huawei Technologies Company Ltd. He is currently an Associate Professor with XDU.